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Field: IT

Final-Year Project Report

CST LePoint System

An intelligent platform for transport management, attendance tracking, and real-time monitoring

Student	Farhani Rayen
Supervisor	Mr Bouslahi Hamdi
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Abstract

This report describes the design and implementation of **CST LePoint**, a full-stack, enterprise-grade platform built for the IT service company **CST (Canadian System Technology)** to digitize and optimize fleet tracking and worker attendance management. The solution solves the real-world operational challenges of manual transport rosters and lack of real-time visibility by offering real-time bus tracking via IoT modems, ge-fenced route-deviation notifications, automated biometric/RFID attendance checking, and interactive business intelligence (BI) layouts.

Furthermore, the system integrates a machine learning microservice built with FastAPI to predict Estimated Time of Arrival (ETA), duration of operations, and absence-risk assessment. Built using a robust technology stack including Angular 19, Tailwind CSS, .NET 8 (following Clean Architecture, CQRS with MediatR, and Carter module routing), and SQL Server, this system enforces multi-tenant security, strict data invariants, and containerized deployment via Docker.

Keywords: Fleet Management, Attendance Tracking, Clean Architecture, CQRS, Machine Learning, FastAPI, Angular, IoT Telemetry.

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List of Acronyms

Acronym	Full Meaning
API	Application Programming Interface
BI	Business Intelligence
CQRS	Command Query Responsibility Segregation
CST	Canadian System Technology
EF Core	Entity Framework Core
ETA	Estimated Time of Arrival
GPS	Global Positioning System
HMAC	Hash-based Message Authentication Code
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
JWT	JSON Web Token
ML	Machine Learning
ORM	Object-Relational Mapping
RBAC	Role-Based Access Control
RFID	Radio-Frequency Identification
REST	Representational State Transfer
SPA	Single Page Application
SQL	Structured Query Language
UI	User Interface
ULID	Universally Unique Lexicographically Sortable Identifier
UML	Unified Modeling Language
WSL	Windows Subsystem for Linux

General Introduction

Over the past decade, the rapid advancement of the Internet of Things (IoT), cloud computing, and artificial intelligence has revolutionized traditional industrial management workflows. In the domain of corporate logistics, transportation systems have transitioned from passive support services into highly integrated, data-driven operations. For large enterprises operating in industrial zones—such as manufacturing plants, automotive assembly facilities, and call centers—organizing the daily transit of thousands of workers across multiple shifts is a complex operational challenge.

Ensuring that employees arrive on time directly impacts production line efficiency, labor relations, and organizational cost control. Traditionally, companies managed these services using fragmented, manual workflows. Roster scheduling was planned on paper or spreadsheets, attendance checks were conducted manually by drivers, and real-time fleet positions were completely invisible. This lack of integration created significant operational bottlenecks: delayed routes went unnoticed, driver identity could not be verified in real time, and invoicing subcontractor fleets was prone to accounting discrepancies.

Report Plan

This thesis documents the complete engineering process of the CST LePoint platform, from requirements analysis to architectural design, implementation, and deployment. The report is structured into five core chapters:

1. **Chapter 1: Project Overview** introduces the host company, Canadian System Technology (CST), details the operational problems, outlines the target user personas, and defines the technical scope and activities of the project.
2. **Chapter 2: Study of Existing Solutions** critiques legacy fleet tracking and biometric gate solutions, evaluates the baseline manual workflows, and presents a concrete operational scenario illustrating the financial impact of transport delays.
3. **Chapter 3: Requirements Specification** defines the functional and non-functional requirements of the platform, presents the use case models (UC-1 through UC-6), and summarizes functional traceability.
4. **Chapter 4: Proposed Solution** details the system design, highlighting Clean Architecture layers, Domain-Driven Design (DDD) concepts, strongly-typed ULID identifiers, database schemas, and MediatR pipeline behaviors.
5. **Chapter 5: Implementation** presents the technology stack, agile Scrum sprint achievements, detailed C# and TypeScript code listings, Docker Compose configuration scripts, and quality assurance testing routines.

Finally, a General Conclusion summarizes the contributions of the project and outlines future extensions for AI-assisted logistics in corporate transport networks.

1 Project Overview

1.1 Introduction

In modern industrial and logistics-intensive organizations, employee transportation has evolved from a simple secondary service into a core operational dependency. For enterprises operating with multiple work shifts, distributed factories, or extensive call center networks, the coordination of staff transit directly impacts productivity, labor satisfaction, and operational budgets. Meeting these demands requires real-time visibility, automated attendance tracking, route compliance auditing, and intelligent planning tools.

The **CST LePoint** platform was initiated to address these challenges by centralizing vehicle fleet management, employee pickup routing, real-time geofenced trace supervision, and introducing an artificial intelligence layer to predict bus Estimated Time of Arrival (ETA) and assess employee absenteeism risks. This chapter establishes the project's background, detailing the host company, the operational problem statement, target user personas, technical objectives, and scope boundaries.

1.2 Host Company

The host company for this project is **Canadian System Technology (CST)**, a pioneering enterprise specializing in Information Technology services, systems integration, and software development. Founded in Montréal, Canada, in 1989 by Mr. Samir Gara, Canadian System Technology established its primary subsidiary in Tunisia in 1997. Over the past three decades, the company has built its reputation on engineering enterprise-grade solutions, consolidating its capabilities through the fusion of four specialized technology entities:

- **GIC:** The developer and publisher of the *Le Point* Enterprise Resource Planning (ERP) suite. The ERP comprises several specialized modules, including:
 - *Le Point-GPAO*: Computer-Aided Production Management (Gestion de Production Assistée par Ordinateur), optimizing industrial manufacturing flows.
 - *Le Point-GPM*: Maintenance Management (Gestion des Processus de Maintenance), tracking equipment lifecycles and interventions.
 - *Le Point-GESTCOM*: Commercial Management (Gestion Commerciale), automating purchasing, sales, and inventory operations.
 - *Le Point-FICO*: Financial and Accounting Management (Gestion Financière et Comptable), handling general ledger, tax filing, and cash flow.
 - *Le Point-GRH*: Human Resources Management (Gestion des Ressources Humaines), managing personnel records, biometric pointage, payroll, and shifts.
- **GIC-Distribution:** An infrastructure systems integrator focusing on high-availability and security solutions. Its key areas of expertise include:
 - Corporate cybersecurity and network access control.
 - Systems consolidation, server virtualization, and high-performance computing clusters.
 - Enterprise storage, backup strategies, and disaster recovery planning.
- **Bs4m:** A specialized publisher of mobile business solutions. Bs4m focuses on bridging physical field activities with centralized enterprise logic, offering technologies in:
 - Mobile Point of Sale (POS) and retail management.
 - Mobile field service intervention and ticketing.
 - Fleet tracking, telemetry, and geolocation services.
- **Canadian Software Technology:** A software development and management consulting firm leveraging cutting-edge web and system frameworks to design custom systems

for production planning, integrated ERPs, and mobile banking.

The **CST LePoint** platform represents a strategic integration within GIC's software ecosystem. By combining the human resource and scheduling principles of GIC-GRH with the real-time telemetry and geofencing capabilities developed by Bs4m, the platform delivers a unified system to manage both personnel attendance and fleet geolocation.

1.3 Project Context and Existing System

1.3.1 Industrial Transport Logistics in Tunisia

Many Tunisian industrial enterprises (such as automotive wire-harness factories, textile plants, and call centers) operate 24/7 across three consecutive 8-hour shifts. Because these factories are often located in industrial zones far from residential districts, companies must organize transport networks to pick up thousands of employees from various pickup points daily. This requires managing dozens of rented or corporate buses, planning optimized paths, and verifying that employees board their designated vehicles.

1.3.2 The Traceability Challenge

Before the initiation of CST LePoint, coordination between dispatchers, drivers, and corporate HR departments relied almost entirely on manual processes, leading to critical inefficiencies. There was no real-time geolocation of buses, attendance checks were performed using manual paper logs, and drivers' routes could not be verified for compliance.

1.3.3 CST LePoint Legacy Platform (Existing Solution)

Before the implementation of the new web-based and intelligent CST LePoint platform, the host company deployed a legacy desktop application to manage transit scheduling. This existing solution suffered from significant limitations:

- **Desktop-Only Access:** The application was a fat-client desktop installation, restricting access to designated office workstations. Dispatchers could not monitor operations on the field or from home, and employees had no visibility into bus locations or schedules.
- **Barebone Operations:** The platform served primarily as a static registry for bus

circuits and driver contacts. Geolocation and route compliance were handled retrospectively, with no real-time tracking or automatic alert systems.

- **No Predictive Analytics:** There was no capability to forecast arrival times (ETA) or identify potential worker absenteeism. It was entirely reactive, depending on phone calls to determine bus delays.
- **Simple Business Intelligence:** Reporting was basic and static, showing only simple aggregate figures without interactive dashboards, custom layouts, or multi-dimensional analytics.

1.3.4 Development Environment and Deployment Stack

To resolve these inefficiencies, our internship project introduces a modernized web-based architecture. A high-level overview of the technological stacks for both systems includes:

- **Legacy System:** A fat-client desktop application directly connected to a centralized database server. Deployment was restricted to local office terminals, and data collection from the field was manually transcribed from paper sheets.
- **New Web Platform:** An Angular 19 frontend Single Page Application (SPA) utilizing Tailwind CSS and Leaflet maps, communicating with a .NET 8 (ASP.NET Core) backend Web API.
- **Machine Learning Service:** A FastAPI Python microservice running alongside the backend to provide predictive analytics (ETA and absenteeism risks).
- **Deployment Stack:** Containerized using Docker Compose to host the frontend (via Nginx), backend API, SQL Server database, and the FastAPI ML service in isolated environments. IoT modems onboard the buses send real-time coordinates and attendance data via the MQTT protocol.

1.4 Project Description

The primary goal of our internship project is to replace the legacy desktop platform with this intelligent, web-based transport and attendance management system. The platform will automate tracking, verify route compliance, and leverage machine learning to provide predictive metrics for dispatchers and commuters.

1.4.1 Target User Personas

The simplified system boundaries involve four core user roles:

- **Administrators:** Manage global configurations, tenant settings, and security roles.
- **Operators:** Monitor active routes, driver assignments, and receive real-time geofence alerts.
- **Analysts:** Generate business intelligence reports, audit telemetry traces, and customize reporting layouts.
- **Commuters (Employees):** Access the web client to view their shift schedules, routes, and live bus positions.

1.4.2 Operational Problems and Risk Boundaries

Developing and deploying a real-time, telemetry-intensive platform introduces key technical and operational risks:

- **GPRS Connection Dropouts:** Loss of cellular coverage in rural zones, requiring local caching and timestamp audits to ensure data integrity.
- **Machine Learning Service Latency:** Delay in predicting ETA, resolved by using asynchronous parallel calls with deterministic fallback rules.
- **Data Ingestion Volume:** Large volume of telemetry coordinates causing database bottlenecks, mitigated by EF Core query splitting and indexing.
- **Multi-Tenancy Security:** Risks of cross-company data leakage, mitigated by enforcing database-level global tenant filters.

1.5 Conclusion

This chapter presented the host company context, the legacy CST LePoint desktop application, the development stack, the deployment architectures for both the legacy and new systems, and the overall objectives of the new project.

However, the legacy system still poses several unresolved operational difficulties. The next chapter, *Study of the Existing*, will further discover the existing solution in detail, analyzing its limitations, the financial and operational impact of those limitations, and the identified rooms for improvement.

2 Study of Existing Solutions

2.1 Introduction

In corporate transport and workforce dispatching logistics, managing fleet assets and tracking personnel presence have historically been treated as distinct administrative concerns. This fragmentation results in operational inefficiencies, security vulnerabilities, and high manual overhead. This chapter reviews the standard commercial systems currently used in the market, analyzes the manual coordination processes replaced by CST LePoint, evaluates a concrete operational failure scenario, and critiques the specific limitations of these legacy solutions.

2.2 Analysis of Commercial Solutions

To evaluate the current state of the art, we analyze three categories of commercial systems commonly deployed in fleet tracking and personnel pointage:

2.2.1 Dedicated Fleet Geolocation Systems (Wialon & GPSTGate)

Wialon and GPSTGate are market-leading commercial platforms designed for vehicle telematics and fleet tracking. They operate by receiving coordinates and diagnostic indicators from physical GPS modems (such as Teltonika or Ruptela devices) installed in vehicles, parsing these packets, and rendering real-time markers on web-based maps.

- **Strengths:** Highly mature cartographic engines, robust support for geofenced speed alerts, fuel level monitoring, and generic travel history reports.
- **Limitations:** These platforms are completely isolated from corporate databases. They operate without any knowledge of who is inside the vehicle. For employee transit, this means dispatchers cannot verify whether the correct operators have boarded, nor can

the system associate driver shifts or track absent workers. The lack of integration with corporate HR directories prevents these systems from solving the employee transit tracking problem.

2.2.2 Biometric Gate Terminals (ZKTime & stationary scanners)

Stationary biometric pointage systems (such as ZKTime terminals, fingerprint turnstiles, and facial recognition devices) are installed at factory gates or main office entrances to audit employee arrival times.

- **Strengths:** High verification accuracy, resistance to buddy-punching (credential sharing), and direct integration with local payroll and HR scheduling modules.
- **Limitations:** These systems suffer from a severe operational blind spot. They only record when an employee crosses the office gate. If a company-provided bus is delayed, is stuck in traffic, or breaks down, the HR system simply flags the employees as late or absent. Dispatchers and plant managers have no way of knowing if the missing workers are on their way, on which vehicle they are riding, or if they are still waiting at their pickup points. This lack of mobility tracking leads to delayed scheduling decisions and production bottlenecks.

2.2.3 Manual Coordination and Communication

For many local small-to-medium enterprises, transportation monitoring is handled using basic communication platforms (such as paper list sheets, phone coordination, and SMS or WhatsApp dispatch groups).

- **Strengths:** Zero capital cost and immediate setup.
- **Limitations:** Manual tracking does not scale. Drivers must manually check off names while operating vehicles, creating safety risks. Paper log sheets are frequently lost, illegible, or subject to falsification. Cross-referencing monthly paper sheets with contractor invoices requires significant administrative labor and is highly prone to calculation errors.

2.3 Current Dependency Tracking Process

The manual tracking workflow replaced by CST LePoint follows five distinct stages:

1. **Weekly Rostering:** Every Friday, HR administrators create employee transport assignments in Excel sheets, linking workers to specific routes and shifts.
2. **Roster Distribution:** Transport coordinators print paper lists and distribute them to drivers.
3. **Manual Boarding Check:** As workers enter the bus, the driver verifies identity card details and manually checks off names.
4. **Status Communication:** In the event of traffic jams or vehicle failures, drivers must call the dispatcher to explain their location verbally.
5. **Invoice Auditing:** At the end of the month, paper rosters are transcribed into spreadsheets. Administrators cross-reference these sheets with invoices from external transit providers to calculate mileage, fuel allowances, and driver wages.

The sequence of operational steps and communication channels for this manual tracking workflow is depicted in Figure 2.1.

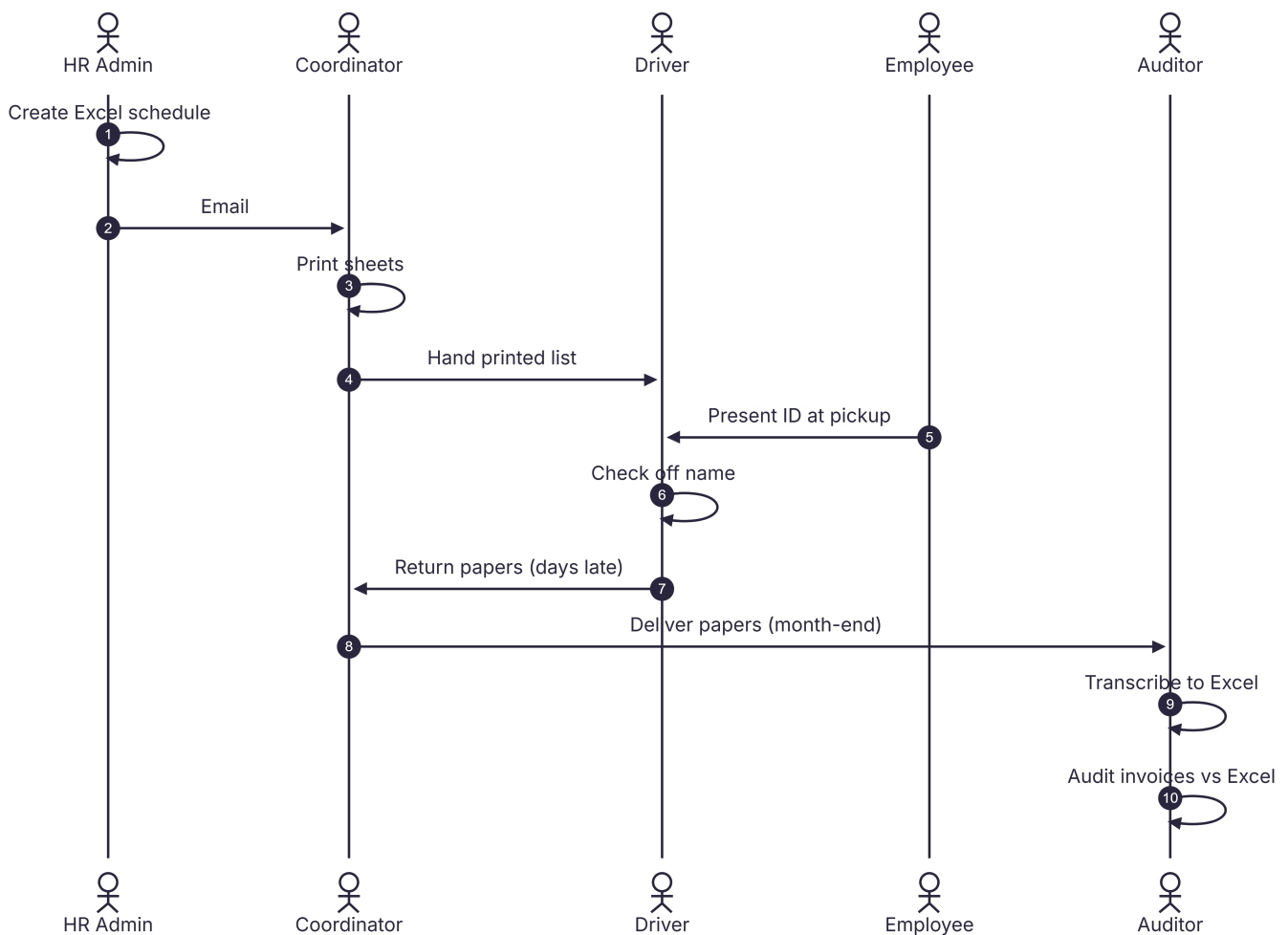


Figure 2.1: Manual Boarding and Attendance Process Sequence

2.3.1 Legacy Deployment Architecture

In the legacy system, the technological architecture was restricted to a centralized database server with local office desktop fat-client instances. Information about transit execution remained isolated in physical logbooks and driver notes, with data manually entered long after the events occurred. Figure 2.2 illustrates the legacy desktop-to-database flow and the lack of real-time network integration with the active fleet.

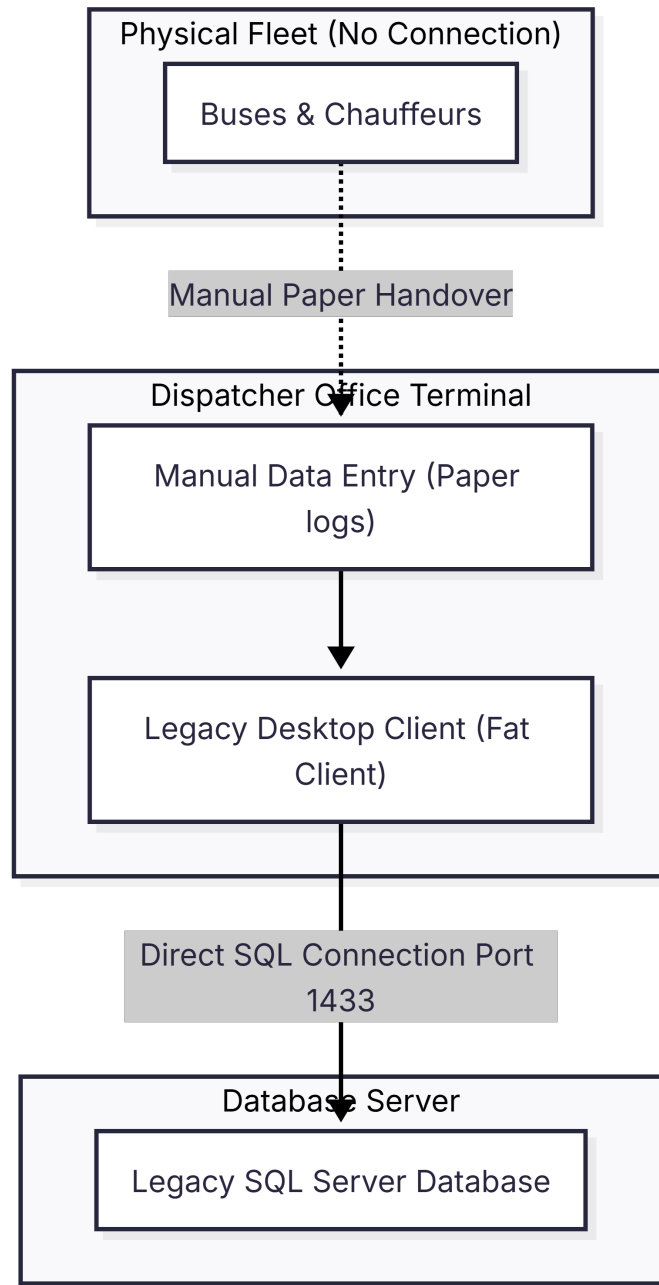


Figure 2.2: Legacy Deployment Architecture and Data Flow

Criteria	Manual book	Log-	GPS Tracking	Office	Biomet-	CST LePoint
				rics		
HR Database Link	None		None	Direct only)	(Gate	Fully Integrated
Live Geolocation	None		Yes	None		Yes + Geofencing
Boarding Audit	Paper sheets		None	None		Biometric/RFID check
Route Audit	Visual inspect		Basic traces	None		Automatic geofencing
Delay Predictions	None		None	None		ML ETA (FastAPI)
Security & RBAC	None		Weak	Strong		Granular RBAC
Operational Cost	High labor		Medium (License)	Low		Low (Automated)

Table 2.1: Comparative Matrix of Transport and Attendance Tracking Solutions

2.4 Financial and Operational Impact of Transport Delays

To understand the necessity of an integrated system, we must evaluate the financial and operational consequences of transit failures in industrial manufacturing environments:

2.4.1 The Assembly Line Vacancy Problem

In automotive parts assembly or electronic manufacturing plants operating on lean manufacturing models, production lines require a strict complement of operators to run safely. If a bus carrying 40 operators is delayed by 30 minutes, the assembly line cannot run at full speed. This leads to immediate bottlenecks, idle workstations downstream, and delayed product shipments.

2.4.2 Concrete Delay Calculation

Consider a standard Tunisian automotive wiring-harness factory with an hourly production value of 15,000 TND.

- A bus breakdown occurs, delaying 40 specialized operators for 1.5 hours.
- Because the operators are missing, a primary production line is halted.
- The financial cost of this halt is calculated as:

$$\text{Loss} = \text{Hourly Production Value} \times \text{Duration} + \text{Overtime Compensations} = 15,000 \times 1.5 + 2,500 = 25,000$$

- In a manual environment, the plant manager is notified only after the shift starts, leaving no time to reassign workers from other lines.

By contrast, an automated real-time tracking system identifies when a vehicle stops moving on a scheduled route, evaluates the geofence breach, calculates the delayed ETA, and alerts dispatchers within 5 minutes, allowing managers to adjust shift patterns proactively.

2.5 Detailed Operational Scenario

Consider a morning shift breakdown scenario: At 6:15 AM, a corporate bus carrying 40 factory operators scheduled for the 7:00 AM shift breaks down on a secondary highway.

- **Roster State:** No live GPS track is active. The driver tries to call the dispatcher, but the line is busy.
- **Search and Rescue:** When the driver finally connects at 6:40 AM, they cannot provide exact coordinates. The dispatcher must call other vehicles to locate the broken bus.
- **Result:** The replacement bus reaches the site at 7:20 AM and delivers the workers to the factory at 7:50 AM, causing a major assembly line shutdown.

The sequence of events and communication breakdowns during this scenario is illustrated in Figure 2.3.

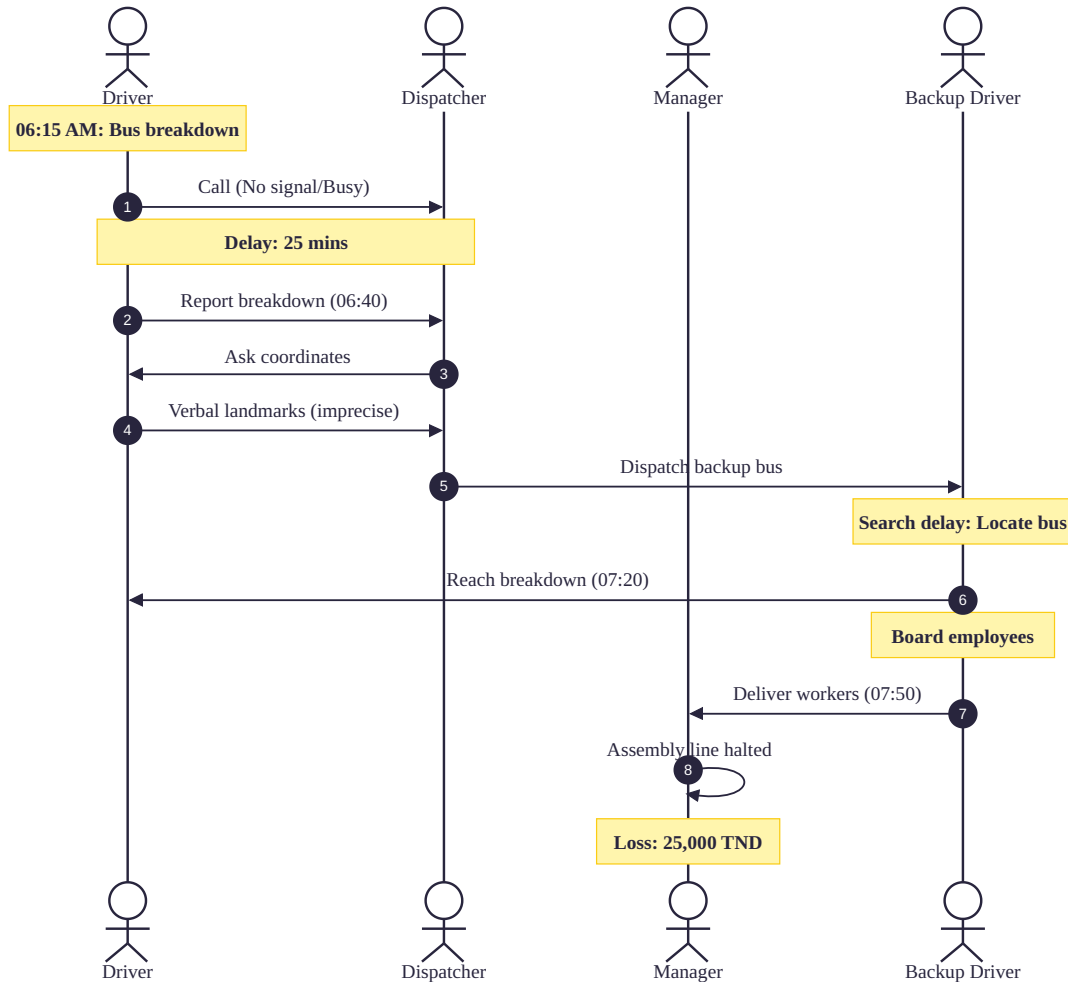


Figure 2.3: Communication and Logistical Timeline during a Vehicle Breakdown

2.6 Limitations and Their Impact

The critique of the manual and fragmented approach highlights several structural, operational, and technical limitations. These constraints directly degrade the logistical efficiency of industrial transit operations, causing significant financial waste, security vulnerabilities, and excessive administrative overhead.

2.6.1 Detailed Technical and Operational Limitations

The legacy system suffered from four fundamental operational and technical limitations:

- **Desktop-Only Constraints:** The legacy application was deployed as a desktop fat-client directly wired to the SQL Server database. This localized setup meant that

transit scheduling was only accessible from office workstations. Field dispatchers could not update schedules or driver details dynamically from the yard, managers had no remote operational visibility, and employees had no way to check bus schedules or routes.

- **Manual Paper Rosters:** Attendance check-ins relied entirely on printed sheets that drivers manually filled during boarding. This method is highly vulnerable to environmental damage (such as water or tearing), illegible handwriting, and data discrepancies. Furthermore, the attendance records remained in the vehicle for days or weeks before being manually transcribed, creating a massive latency in HR record updates.
- **Lack of Real-Time Geolocation:** With no real-time telemetry, the dispatcher office was completely blind to vehicle positions. The system served as a static database of planned circuits and driver contacts. If a bus was delayed, coordinators could only estimate its location and status by making direct phone calls to the driver, which is both dangerous while driving and prone to subjective reports.
- **Absence of Route Compliance Auditing:** There was no telemetry feedback loop to verify if drivers followed the scheduled circuits. Drivers could skip pickup points, take unauthorized detours, or deviate from planned paths to avoid traffic without the system registering any anomaly. This lack of auditing made it impossible to enforce route compliance or optimize transit loops.

2.6.2 Business and Financial Impact on Logistics

These technical limitations translate directly into measurable business and financial losses for the company:

- **Delayed Shifts and Assembly Line Vacancy:** Tunisian manufacturing plants, such as automotive wiring-harness factories, operate on strict lean production models where workstations are tightly sequenced. A delay of a single bus carrying 30 or 40 technicians results in vacant assembly stations, slowing or halting the entire downstream line. The financial cost of such delays is severe, as calculated in the breakdown scenario (25,000 TND in idle labor and overtime recovery costs for a 1.5-hour delay).
- **Fuel Wastage and Unmonitored Mileage:** In the absence of automated route auditing, external transport contractors frequently bill for routes that are longer than necessary or routes they did not fully complete. Excessive engine idling, detours for driver personal errands, and unoptimized paths cause substantial fuel wastage and inflated subcontractor billing.

- **Security and Liability Risks:** Manual rosters do not offer secure verification of passengers. If a traffic accident or roadside emergency occurs, the company cannot quickly determine which employees were aboard the vehicle, creating serious regulatory and safety compliance liabilities. Furthermore, paper rosters are insecure and can easily be accessed by unauthorized personnel, exposing sensitive employee schedules and locations.
- **Administrative Burden and Invoice Auditing Delay:** At the end of every monthly billing cycle, administrative teams must manually type thousands of paper check-in entries into Excel files and cross-reference them with subcontractor invoices. This manual verification takes dozens of hours, is highly prone to human mathematical errors, and delays payments to transit providers.

2.7 Identified Rooms for Improvement and Transition

To resolve these operational difficulties and secure its logistics chain, Canadian System Technology requires a modernized, unified platform. The transition from the legacy desktop client to a cloud-ready web and mobile framework must focus on three core areas of improvement:

1. **Real-Time Telemetry and Cartographic Supervision:** Replace static database registries with an active, web-based map interface. Using IoT modems installed in the buses, the system must process real-time coordinates via standard protocols (such as MQTT or HTTP) and dynamically plot vehicle positions, allowing dispatchers to monitor route progress and catch delays instantly.
2. **Automated Attendance Tracking:** Replace manual paper sheets with digital boarding checks using RFID card scanners linked directly to the on-board IoT modems. As employees tap their company cards, the system must instantly validate their credentials, log their boarding status, and update the centralized HR database in real time.
3. **Predictive Analytics and Intelligence:** Incorporate a machine learning layer to move from reactive scheduling to proactive management. The system should automatically query historical transit data and traffic conditions to calculate accurate Estimated Times of Arrival (ETA) for commuters, and analyze attendance traces to predict absenteeism risks before shifts begin.

2.8 Conclusion

This chapter detailed the existing system and the limitations of manual transportation tracking workflows. It demonstrated the operational and financial impact of delays on lean manufacturing lines and identified key areas for improvement, establishing that a web-based, real-time, and predictive system is necessary to replace the legacy fat-client.

To implement this modernized system, a comprehensive set of specifications must be defined. The next chapter establishes the functional and non-functional requirements specification for the proposed CST LePoint platform.

3 Requirements Specification

3.1 Introduction

Defining precise requirements is essential to building an enterprise-grade solution that aligns technical implementations with operational realities. This chapter details the functional and non-functional requirements, presents the global and sub-system use case models, and establishes the boundaries of the CST LePoint system.

3.2 Functional Requirements

The functional requirements cover all the actions that the system must support. To align these requirements with modern software engineering practices, they are organized into core operational modules and detailed as User Stories:

3.2.1 Authentication and Authorization

- **FR-01: Secure Sign-In** (Sprint 1, 3 SP, Priority: High)
As a User, I want to authenticate with credentials and SocieteId so that I can access my company scope.
- **FR-02: Stateless Token Generation** (Sprint 1, 2 SP, Priority: High)
As a User, I want to receive a signed JWT token upon login so that my session is authenticated without server-side state.
- **FR-03: Token Revalidation** (Sprint 1, 1 SP, Priority: Medium)
As a Client App, I want to verify the token validity on startup so that I can automatically restore active sessions.
- **FR-04: Granular Access Control** (Sprint 2, 5 SP, Priority: High)
As an Administrator, I want to assign navigation and API permissions to roles so that

users only see and execute authorized actions.

3.2.2 Fleet Management

- **FR-05: Bus Registration** (Sprint 2, 2 SP, Priority: High)

As an Administrator, I want to manage buses (immatriculation, model, capacity) so that they can be assigned to routes.

- **FR-06: Driver Assignment** (Sprint 2, 2 SP, Priority: High)

As an Operator, I want to link drivers to buses and RFID badges so that operational attendance can be registered.

- **FR-07: Modem Association** (Sprint 2, 2 SP, Priority: Medium)

As an Administrator, I want to catalog IoT modems and pair them with buses so that telemetry can be mapped to active vehicles.

3.2.3 Pickup Planning

- **FR-08: Circuit Definition** (Sprint 2, 3 SP, Priority: High)

As an Operator, I want to create circuits with GPS coordinates, distance, and duration so that routes can be planned.

- **FR-09: Ordered Pickup Points** (Sprint 2, 3 SP, Priority: High)

As an Operator, I want to sequence pickup points along a circuit so that the bus follows a strict logistical route.

3.2.4 Passenger and Shift Management

- **FR-10: Employee Directory** (Sprint 2, 3 SP, Priority: High)

As an Administrator, I want to manage employee records, RFID tags, and geographic coordinates so that pickups can be targeted.

- **FR-11: Shift Configuration** (Sprint 2, 2 SP, Priority: Medium)

As an Administrator, I want to define working shifts (start/end times) so that employees can be scheduled on transit routes.

- **FR-12: Operational Work Orders** (Sprint 2, 3 SP, Priority: Medium)

As an Operator, I want to schedule site work orders for teams so that logistical transport matches factory demand.

3.2.5 IoT Telemetry Ingestion

- **FR-13: Real-Time Telemetry Ingestion** (Sprint 4, 5 SP, Priority: High)
As a Modem, I want to submit periodic GPS/RFID occupancy payloads to the API so that fleet status is updated in real-time.
- **FR-14: Telemetry Replay Protection** (Sprint 4, 2 SP, Priority: High)
As a System, I want to reject telemetry payloads older than 15 minutes so that replay attacks are prevented.
- **FR-15: Automatic Geofence Tracking** (Sprint 4, 5 SP, Priority: High)
As a System, I want to flag a bus detour if it deviates more than 250m from the route so that deviations are logged immediately.

3.2.6 Real-Time Supervision

- **FR-16: Cartographic Tracking** (Sprint 3, 5 SP, Priority: High)
As an Operator, I want to view active buses moving in real-time on a map so that I can supervise fleet routes visually.
- **FR-17: Telemetry Timeline** (Sprint 3, 3 SP, Priority: Medium)
As an Operator, I want to see a chronological log of events for each bus so that I can analyze runtime alerts and updates.

3.2.7 Predictive Analytics

- **FR-18: ETA Forecasting** (Sprint 4, 5 SP, Priority: High)
As an Operator, I want to receive machine-learning predicted ETAs for active buses so that I can inform plants of potential delays.
- **FR-19: Batch ETA Calculations** (Sprint 4, 3 SP, Priority: High)
As a Backend, I want to query FastAPI for ETAs of all active buses in parallel so that tracking updates remain efficient.

3.2.8 Business Intelligence Reporting

- **FR-20: Dynamic Querying & Reporting** (Sprint 5, 5 SP, Priority: High)
As an Operator, I want to run historical telemetry and attendance reports with sorting, grouping, and exporting so that I can audit transport operations.

- **FR-21: Custom Report Layouts** (Sprint 5, 3 SP, Priority: Medium)

As an Operator, I want to save my custom column groupings and configurations so that I can reload them with one click.

3.3 Non-Functional Requirements

The non-functional requirements describe the quality attributes, technological standards, and operational constraints achieved in the CST LePoint project:

3.3.1 Performance and Ingestion Latency

To handle high-frequency data generated by the active fleet, the system must adhere to strict performance targets:

- **Ingestion Response Time:** Telemetry ingestion API requests (`POST /cm/bus/runtime/positions`) must be processed and resolved in under 200ms to avoid network queues on cellular IoT devices.
- **High Concurrency Support:** The backend must support concurrent telemetry updates from 50+ IoT modems broadcasting GPS/RFID packet data every 30 seconds without database deadlock or memory leaks.
- **UI Grid Load Latency:** Historical datagrids rendering thousands of telemetry trace records must complete client-side sorting, pagination, and filtering in less than 500ms using virtual scrolling and optimized EF Core query parameters.

3.3.2 Scalability and Containerization Boundary

The system architecture must support horizontal scaling as the company's fleet size grows:

- **Container Separation:** Services must be isolated into independent container layers (Frontend client, .NET Core WebAPI, SQL Server 2019, and FastAPI ML prediction service) orchestrated using Docker Compose.
- **Stateless API Layers:** The backend application is stateless, allowing multiple instances of the .NET WebAPI container to run behind a reverse proxy (Nginx) to distribute concurrent REST traffic.

3.3.3 Data Integrity and CQRS Transactional Boundaries

Operational logs and financial reports demand absolute data consistency:

- **Unit of Work Pattern:** Database mutations must be bound by transactional boundaries managed via MediatR Pipeline Behaviors. A failure in any handler step must trigger an automatic transaction rollback.
- **Fail-Fast Input Validation:** Before committing modifications to the persistence layer, incoming data must pass strict FluentValidation checks at the application commands boundary.

3.3.4 Security, Cryptography, and Privacy Standards

As an enterprise-grade platform, security controls are configured at both transmission and storage levels:

- **Stateless Authentication:** All client-server communication must be authenticated using JSON Web Tokens (JWT) signed via HMAC-SHA256, carrying expiration times and client metadata claims.
- **Credential Hashing:** Passwords must never be stored in plain text. They are hashed using BCrypt, utilizing a configurable work factor to protect against brute-force attacks.
- **Action-Level Authorization:** System operations are restricted based on user role permissions (Read, Add, Edit, Delete) mapped dynamically to application routing modules.

3.3.5 System Maintainability and DDD Architectural Styles

To ensure long-term code quality and reduce technical debt:

- **Clean Architecture Rules:** Core business logic (Domain and Application layers) must remain pure, with zero dependencies on external frameworks, SGBDs, or third-party mapping libraries.
- **Domain Invariants Enforcements:** Domain models must use private setters, requiring state changes to flow through explicit constructor factories and validation methods.
- **ULID Indexing:** Distributed, lexicographically sortable ULIDs (Universally Unique Lexicographically Sortable Identifiers) are used as primary database keys to optimize

clustering indexing over classic random GUIDs.

3.3.6 Robustness and Fault Tolerance

The platform must degrade gracefully when external dependencies fail:

- **ML Service Failure Handling:** If the FastAPI machine learning container becomes unreachable, the WebAPI backend must fallback to static duration heuristics based on deterministic circuit distances, logging a warning while keeping the user experience uninterrupted.
- **Replay Attack Defenses:** The ingestion engine must validate telemetry packet timestamps, rejecting any payload delayed by more than 15 minutes to protect against data injection.

3.4 Use Case Model

To model the interactions between users, external systems, and the CST LePoint platform, we establish a formal use case model. We identify the primary actors and external systems, present the global use case diagram, detail the sub-system use case models, and specify the operational flows of core use cases.

3.4.1 Identification of Actors and External Systems

To maintain a clean and coherent system boundary, the human interactions with the platform are structured around two primary roles, complemented by two external automated systems:

Human Actors

- **Administrator:** Manages company tenant settings, user accounts, dynamic role-based access control (RBAC) permission matrices, fleet vehicle records, and hardware device registries.
- **Operator / Dispatcher:** Coordinates daily transit operations, including circuit and pickup stop planning, employee pickup assignments (*Rattachements*), shift work orders, live cartographic tracking, arrival time forecasting, and business intelligence reporting.

External Systems

- **IoT Telemetry Modem:** An onboard vehicular telematics unit (GPS/GPRS and RFID scanner) that transmits periodic position packets, timestamps, and passenger occupancy data to the backend ingestion API.
- **ML Prediction Engine:** A specialized FastAPI microservice that computes real-time Estimated Time of Arrival (ETA) predictions based on historical transit trajectories and current bus telemetry.

3.4.2 Global Use Case Diagram

The global use case diagram illustrates the high-level functional boundaries of the CST LePoint platform, organizing system capabilities into three logical packages: *Administration & Security*, *Fleet & Route Logistics*, and *Real-Time Supervision & BI*.

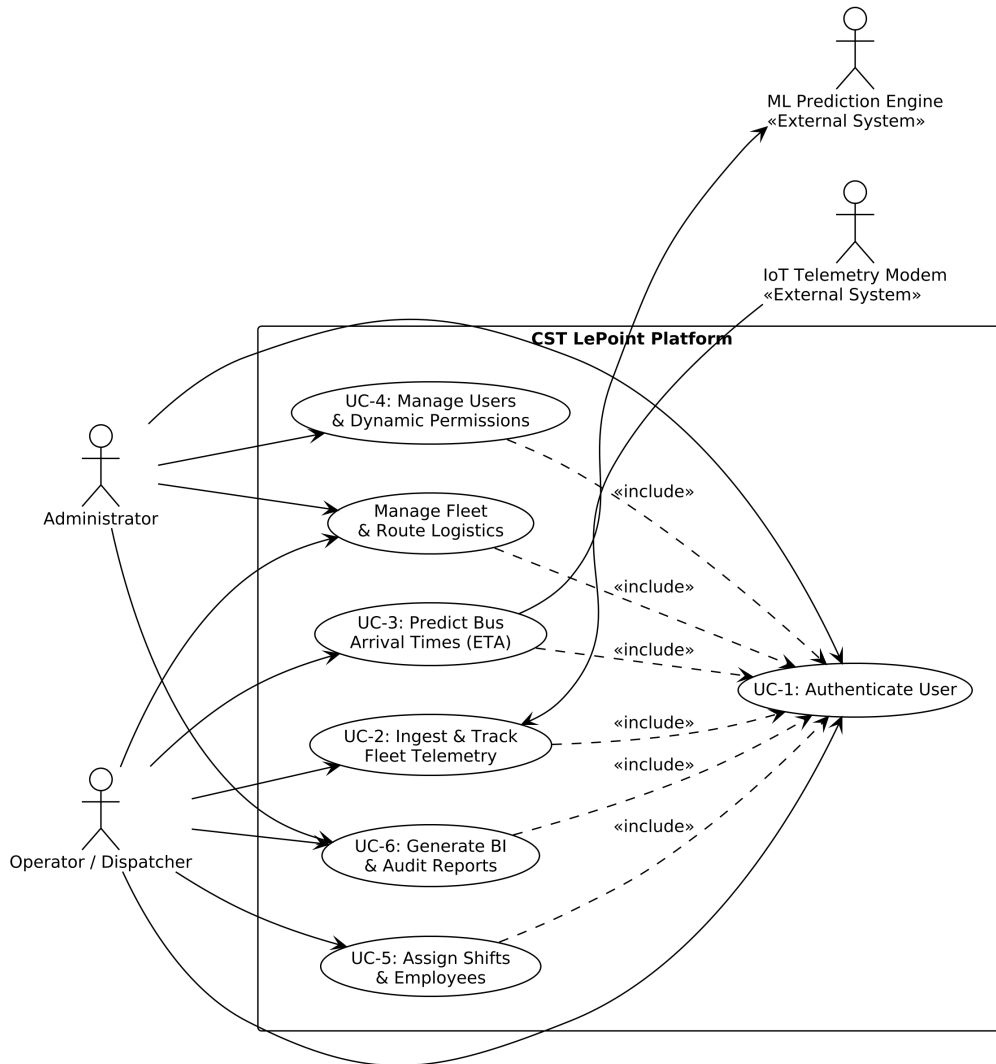


Figure 3.1: Global Use Case Diagram – CST LePoint Platform

3.4.3 Sub-System Use Case Models

To provide granular visibility into specific functional areas, the system scope is decomposed into four focused sub-systems.

Fleet and Route Logistics Sub-System

This sub-system covers the management of physical vehicles, IoT modems, transit circuits, ordered pickup stations, and route allocations. Figure 3.2 depicts the interactions between the Administrator and Operator in configuring transport assets.

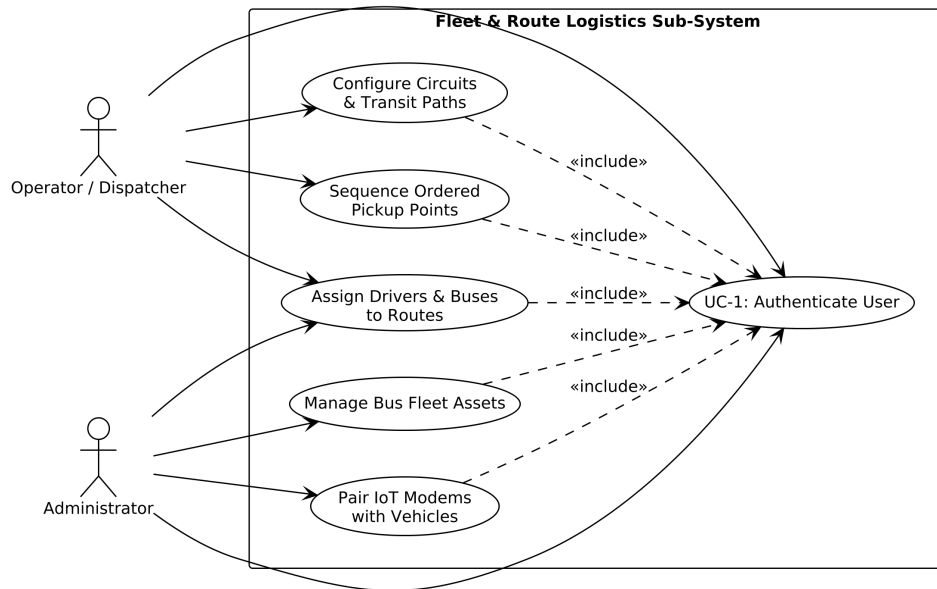


Figure 3.2: Sub-System Use Case – Fleet and Route Logistics

Real-Time Supervision and Telemetry Sub-System

This sub-system captures high-frequency location streaming from vehicular IoT modems, evaluates geofence boundaries, renders live cartographic markers for operators, and requests ETA predictions from the machine learning engine. Figure 3.3 illustrates these interactions.

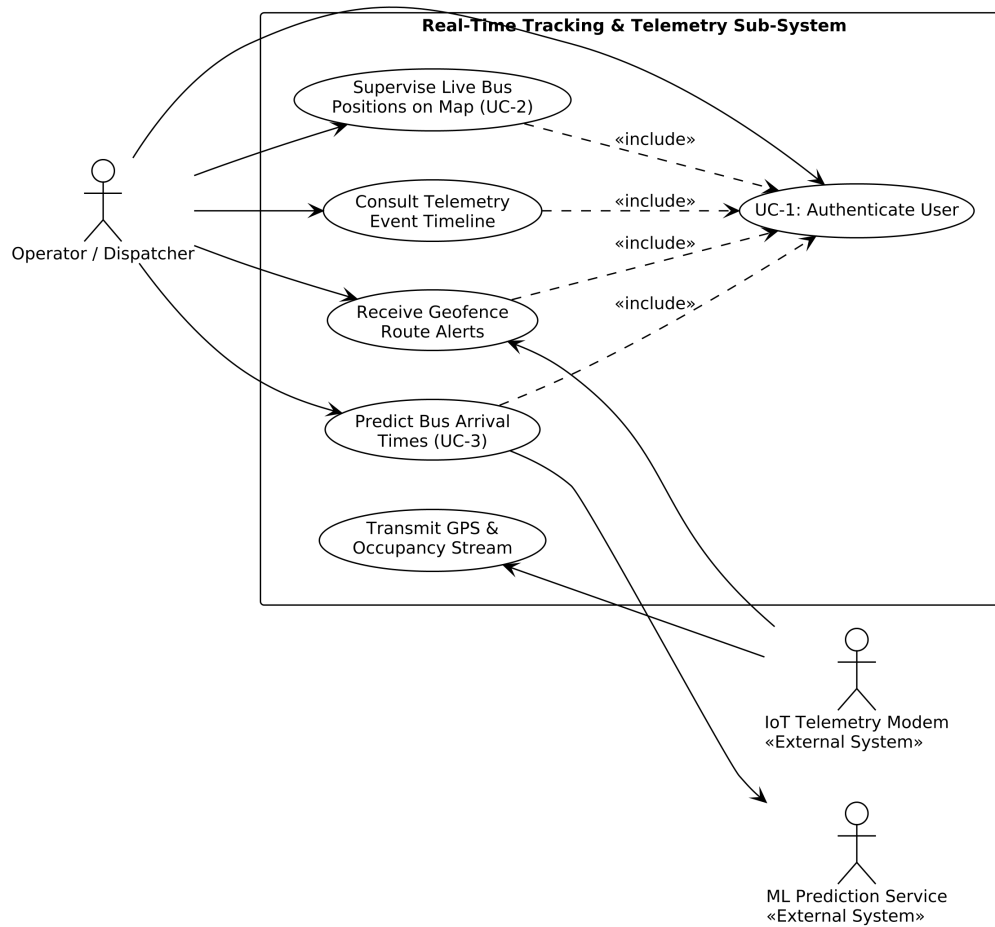


Figure 3.3: Sub-System Use Case – Real-Time Supervision and Telemetry

Operations and Workforce Logistics Sub-System

This sub-system models employee roster assignments, work order scheduling across factory shifts, and historical business intelligence reporting with customizable layout persistence. Figure 3.4 presents the operational model.

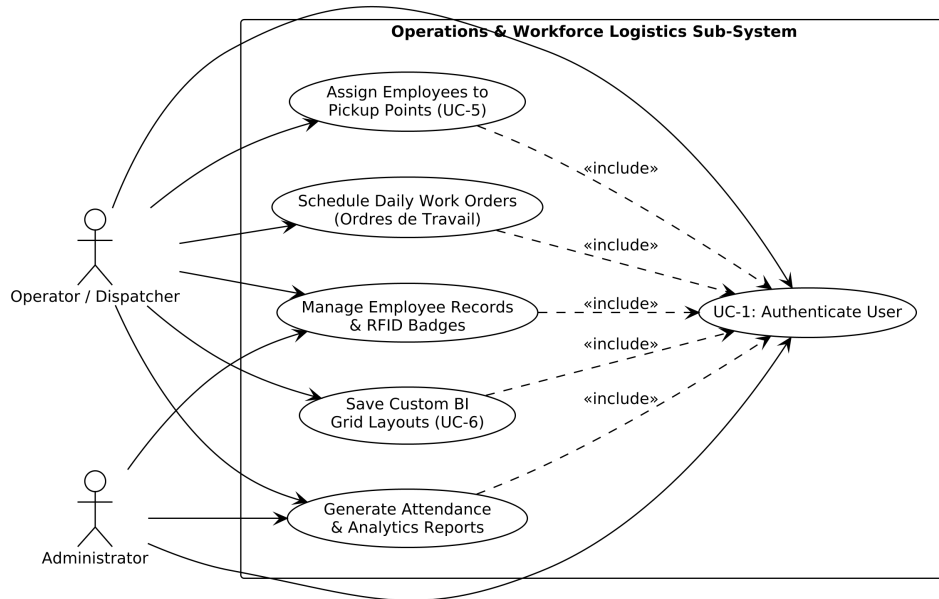


Figure 3.4: Sub-System Use Case – Operations and Workforce Logistics

System Administration and Access Governance Sub-System

This sub-system ensures tenant isolation, user account lifecycle administration, dynamic RBAC permission matrix configuration, and reference catalog maintenance. Figure 3.5 illustrates the administrator workflows.

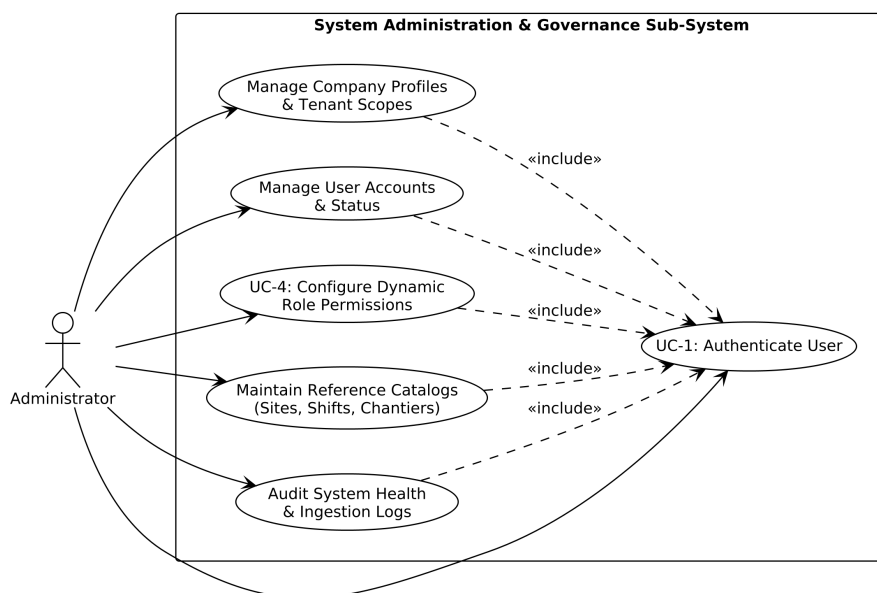


Figure 3.5: Sub-System Use Case – System Administration and Governance

3.4.4 Detailed Use Case Specifications

This section provides formal specifications for key operational use cases developed across the platform:

UC-1: Authenticate User

- **Primary Actors:** Administrator, Operator.
- **Preconditions:** The user account is active, and the client application is loaded at the login screen.
- **Postconditions:** The user session is initialized, a cryptographically signed JWT token is stored, and authorized navigation menus are displayed.
- **Main Flow:**
 1. The user enters their username, password, and company identifier (`SocieteId`).
 2. The user submits the login form.
 3. The client application performs local validation and dispatches an HTTP POST request to `/cm/auth/login`.
 4. The WebAPI receives the payload and passes a login command to the MediatR pipeline.
 5. The command handler retrieves the user entity and verifies the password hash via BCrypt.
 6. The handler resolves the user's role and associated navigation-to-action permission matrix.
 7. The system generates an HMAC-SHA256 signed JWT Bearer token carrying identity and permission claims.
 8. The API returns an HTTP 200 OK response wrapping the token and profile metadata.
 9. The frontend stores the token and restores the user's authenticated state.
 10. The application redirects the user to the operational dashboard.
- **Alternative Flows:**
 - *Invalid Credentials:* If the username or password fails verification, the API returns HTTP 401 Unauthorized, and an error notification is shown.
 - *Inactive Account:* If the account is deactivated (`IsActive = false`), the API returns HTTP 403 Forbidden.

UC-2: Ingest and Track Fleet Telemetry

- **Primary Actors / Systems:** IoT Telemetry Modem, Operator.
- **Preconditions:** The vehicle is active, the onboard modem is initialized, and the operator has opened the live map view.
- **Postconditions:** Vehicle coordinates and passenger occupancy counts are committed to persistence, and live map markers are updated.
- **Main Flow:**
 1. The onboard modem captures GPS coordinates (latitude, longitude, speed) and RFID passenger counts.
 2. The modem transmits an HTTP POST telemetry packet to `/cm/bus/runtime/positions`.
 3. The WebAPI verifies that the packet timestamp falls within a 15-minute replay-protection window.
 4. The system resolves the bus associated with the modem's 15-digit IMEI.
 5. The command handler calculates the spatial distance between the vehicle position and scheduled circuit points.
 6. The handler updates the bus runtime state and persists the telemetry record.
 7. The backend broadcasts the coordinate update via WebSocket to connected clients.
 8. The Leaflet map component updates the vehicle marker position smoothly.
- **Alternative Flows:**
 - *Geofence Route Deviation:* If the bus deviates more than 250 meters from the scheduled circuit path, the system logs a detour alert and generates an event record before committing.
 - *Unrecognized Modem:* If the IMEI is unregistered, the request is rejected with HTTP 404 Not Found.

UC-3: Predict Bus Arrival Time (ETA)

- **Primary Actors / Systems:** Operator / Dispatcher, ML Prediction Engine.
- **Preconditions:** The operator is logged in, and an active bus is in transit along a scheduled circuit.
- **Postconditions:** The machine learning estimated arrival time (in minutes) and confidence metrics are displayed on the tracking panel.

- **Main Flow:**

1. The operator clicks on an active bus marker on the live tracking map.
2. The client issues an ETA prediction query to the backend API.
3. The backend aggregates real-time bus telemetry (speed, current coordinates, passenger occupancy, capacity, and circuit ID).
4. The backend issues an HTTP POST prediction request to the FastAPI microservice (`http://ml-service:8001/predict/eta`).
5. The FastAPI microservice loads the trained regression model artifacts, executes inference, and returns the predicted ETA duration in minutes.
6. The backend WebAPI receives the prediction response and formats it into the API contract.
7. The tracking panel displays the remaining arrival time and estimated arrival timestamp.

- **Alternative Flows:**

- *ML Service Unavailable*: If the FastAPI container is unreachable, the backend logs a warning and falls back to a deterministic distance-to-speed heuristic calculation, ensuring uninterrupted user experience.

UC-4: Configure Role Permissions Matrix

- **Primary Actors:** Administrator.

- **Preconditions:** The Administrator is authenticated and navigates to the Roles and Permissions governance view.

- **Postconditions:** Role permissions are saved in the database as a structured JSON permission matrix.

- **Main Flow:**

1. The Administrator opens the role management interface.
2. The system displays the list of roles and the module permission matrix.
3. The Administrator selects a role to edit or creates a new role.
4. The Administrator configures module privileges (*Read, Add, Edit, Delete*) across functional routes (e.g. `fichier.circuit`, `monitoring.bus-tracking`, `analyse.bi-bus`).
5. The Administrator clicks "Save Role".

6. The client compiles the selections into a JSON array and dispatches an HTTP POST/PATCH request.
 7. The backend handler validates uniqueness within the company tenant and persists the `RoleUtilisateur` entity.
 8. The UI displays a confirmation notification.
- **Alternative Flows:**
 - *Duplicate Role*: If the role name already exists within the same tenant, the API aborts with a validation error.

UC-5: Assign Employee to Pickup Point and Route

- **Primary Actors**: Operator / Dispatcher.
- **Preconditions**: The employee record exists with registered geographic coordinates, and the transit circuit is configured.
- **Postconditions**: An assignment link (`Rattachement`) is created, binding the employee to a designated pickup point and scheduled shift.
- **Main Flow**:
 1. The operator opens the employee pickup assignment module.
 2. The system renders the pickup points map alongside unassigned employee profiles.
 3. The operator selects an employee and identifies the nearest pickup station along an active circuit.
 4. The operator assigns the employee to the chosen pickup point and associated shift.
 5. The client sends an HTTP POST request to create the `RattachementEmploye` entity.
 6. The backend verifies vehicle capacity constraints and commits the assignment record.
 7. The map highlights the updated passenger count for that pickup station.
- **Alternative Flows**:
 - *Overcapacity Warning*: If adding the employee exceeds the bus seating capacity, the system displays a capacity warning prompt before confirmation.

UC-6: Save Custom BI Report Layout

- **Primary Actors:** Operator / Dispatcher, Administrator.
- **Preconditions:** The user is authenticated and has customized a BI data grid (reordered columns, applied filters, or established groupings).
- **Postconditions:** The customized grid configuration JSON is persisted in the database and linked to the tenant context.
- **Main Flow:**
 1. The user opens a business intelligence dashboard (e.g. BI Bus or BI Attendance).
 2. The user customizes grid columns, sorting criteria, and multi-level groupings.
 3. The user clicks "Save Layout".
 4. A dialog prompts for a descriptive layout title (e.g. "Monthly Overtime Audit").
 5. The user enters the name and submits the dialog.
 6. The client serializes the grid state to JSON and sends an HTTP POST request to `/cm/analyse/layouts`.
 7. The backend handler upserts the `ReportLayout` aggregate, scoped by `SocieteId`.
 8. The saved layout appears in the user's layout dropdown menu for instant recall.
- **Alternative Flows:**
 - *Blank Layout Name:* If the user leaves the title blank, the client displays an inline validation error.

3.5 Input Validation Constraints

To maintain data integrity and prevent corrupt persistence writes, the CST LePoint platform enforces strict validation rules at the application boundary using `FluentValidation`. Inbound requests are checked against the following constraints:

- **Bus Entity Constraints:**
 - `NumeroIMM` (Immatriculation plate format): Required, must not exceed 50 characters, and must follow standard Tunisian plate expressions (regex: `^[0-9]+-TN-[0-9]+$`).
 - `IMEI` (Onboard modem identifier): Required, must be exactly 15 numeric digits.
 - `Capacite` (Vehicle seating capacity): Required, must be an integer between 1 and 150.

- **Employee Entity Constraints:**

- **Matricule** (HR employee identification): Required, unique, and cannot exceed 20 characters.
- **RFID** (Badge hardware identifier): Required, must be a valid alphanumeric hexadecimal token up to 50 characters.
- **Latitude/Longitude** (Home geolocation): Must fall within the geographic bounds of Tunisia (Latitude: [30.0, 38.0], Longitude: [7.0, 12.0]).

- **Circuit Entity Constraints:**

- **CodeCircuit** (Route catalog identifier): Required, unique, alphanumeric code up to 20 characters.
- **DistanceKm** (Transit path length): Required, must be a positive decimal value greater than 0.0.
- **DureeMinutes** (Estimated transit duration): Required, must be a positive integer greater than 0.

3.6 Requirements Traceability Summary

To ensure complete coverage of the design scope, we cross-reference our functional requirements with the corresponding use cases and implementation features in Table 3.1:

ID	Requirement Name	Traceable Use Cases / Implementation Components
REQ-01	User Authentication	UC-1, BCrypt Hashing, JWT Bearer Token validation
REQ-02	Navigation Permissions	UC-4, RoleUtilisateur navigation matrices, Navigation Guards
REQ-03	Fleet CRUD Catalogs	Admin dashboard, DevExtreme DataGrids, EF Core context
REQ-04	Telemetry Ingestion	UC-2, minimal API telemetry endpoint, IMEI validation
REQ-05	Live Geofencing alerts	Haversine distance tracking, BusRuntimeEvent logging
REQ-06	Cartographic Map View	ngx-leaflet component, WebSocket polling, marker states
REQ-07	Machine Learning ETA	UC-3, Python FastAPI microservice, PredictBusEta queries
REQ-08	Custom BI Layouts	UC-6, ReportLayout entity, UpsertReportLayout, ApexCharts
REQ-09	Workforce Pickup Assignment	UC-5, RattachementEmploye entity, Capacity validation

Table 3.1: Functional Requirements Traceability Matrix

3.7 Conclusion

This chapter presented the detailed functional and non-functional requirements of the CST LePoint platform, defined the streamlined actor hierarchy, provided global and sub-system use case diagrams, specified core operational flows, and mapped these requirements to implementation components. Having established the system specifications, the next chapter presents the proposed system architecture, database modeling, and logical design solutions.

4 Proposed Solution

4.1 Introduction

Following the comprehensive analysis of existing transit workflows and the detailed operational requirements established in the preceding chapters, this chapter presents the architectural and conceptual design of the CST LePoint platform.

The primary objective of the proposed solution is to transform traditional, fragmented employee transportation and attendance tracking into an integrated, real-time, automated, and intelligent supervision platform. The system is designed to provide end-to-end operational visibility: from master planning and workforce assignment to real-time telemetry streaming, automated route deviation detection, predictive arrival forecasting, and customizable business intelligence reporting.

In this chapter, we focus on the high-level architecture, functional subsystems, information models, and operational workflows that together fulfill the platform’s core mission. The detailed technology stack, source code implementations, and concrete user interface realizations will be presented in Chapter 5.

The chapter is organized around the following key dimensions:

- **High-Level System Architecture and Operational Lifecycle:** The global multi-tier ecosystem, detailed logical Clean Architecture layers, physical containerized deployment topology, end-to-end operational journey, and inter-service communication protocols;
- **Multi-Tenant Identity and Access Governance:** Organizational data isolation and dynamic role-based access control;
- **Fleet and Transit Logistics Planning Subsystem:** Vehicle catalogs, circuit definitions, workforce shift scheduling, and the conceptual domain model;
- **Real-Time Telemetry Processing and Automated Supervision:** High-throughput coordinate ingestion, live cartographic monitoring, and automatic geofence detection;

- **Predictive Intelligence and Transit Forecasting:** Real-time arrival estimation, passenger consultation, and fault-tolerant fallback strategies;
- **Operational Analytics and Business Intelligence Subsystem:** Performance metrics, custom reporting layouts, and data reconciliation;
- **Architectural Cohesion and System Integrity:** High-level architectural foundations ensuring maintainability, separation of concerns, and auditability.

4.2 High-Level System Architecture and Operational Design

4.2.1 Global Multi-Tier Ecosystem

To satisfy the demanding requirements of real-time transit supervision, multi-tenant enterprise usage, and predictive analytics, the CST LePoint platform is architected as a decoupled, multi-tier distributed ecosystem. Each tier addresses a distinct operational and technical concern, communicating over standardized network protocols to ensure loose coupling, high responsiveness, and independent horizontal scalability.

Figure 4.1 illustrates the global multi-tier ecosystem and the primary interaction pathways between its constituent layers.

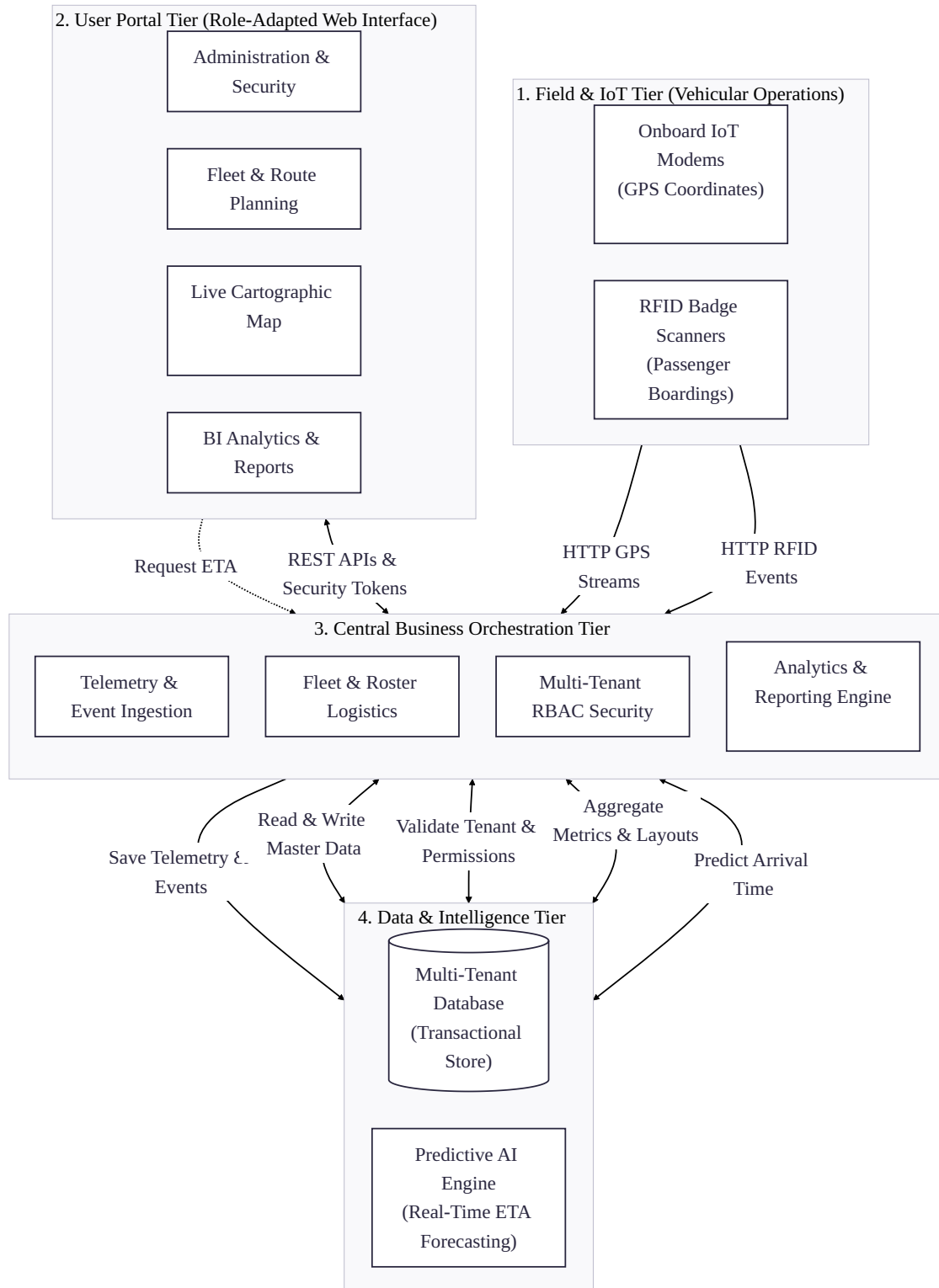


Figure 4.1: Global System Architecture – Multi-Tier Functional Ecosystem

The four architectural tiers operate collaboratively as follows:

1. **Field and IoT Tier:** Comprises embedded hardware units deployed across the transit

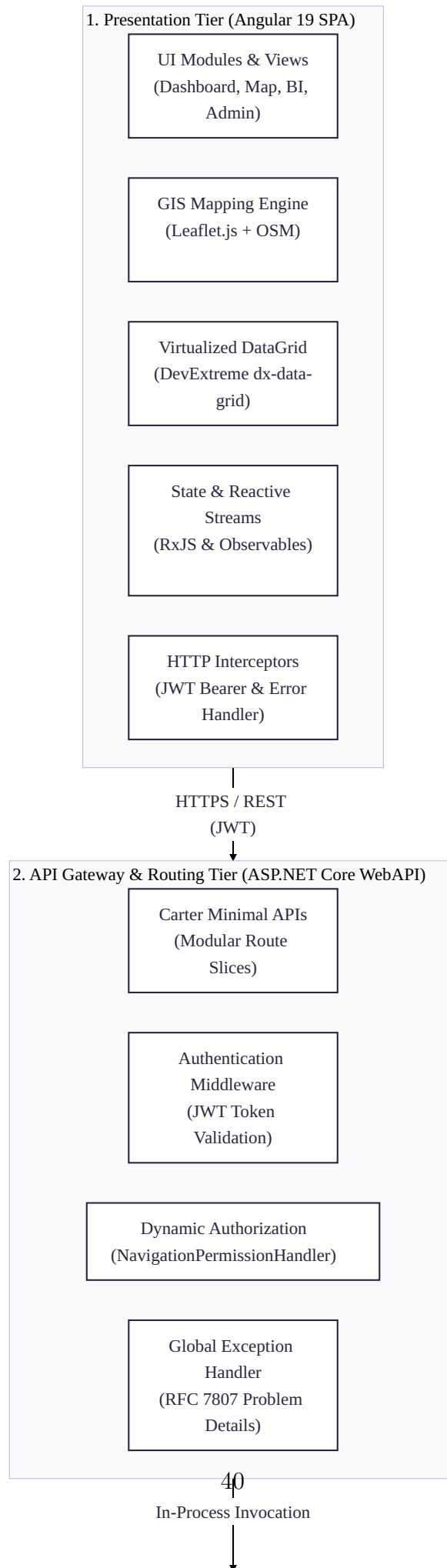
fleet, including GPS modems and RFID badge scanners. These units capture and transmit high-frequency location vectors, speed metrics, and passenger boarding events directly from active vehicles over cellular networks.

2. **User Portal Tier:** Delivers a responsive Single Page Application (SPA) accessible through modern web browsers. Tailored to distinct enterprise user profiles (system administrators, fleet dispatchers, operations supervisors, and commuters), it delivers interactive cartographic tracking, master data planning interfaces, and dynamic analytical dashboards.
3. **Central Business Orchestration Tier:** Serves as the operational brain of the platform. It executes business workflows, validates domain invariants, enforces multi-tenant security boundaries, performs real-time geofence calculations, coordinates transactional persistence, and delegates machine learning inferences.
4. **Data and Intelligence Tier:** Combines a centralized relational database ensuring strict ACID compliance, auditability, and multi-tenant partitioning, alongside a specialized Python artificial intelligence microservice dedicated to real-time bus arrival forecasting.

4.2.2 Logical System Architecture

The internal design of the CST LePoint platform is governed by the principles of Clean Architecture, Domain-Driven Design (DDD), and Command Query Responsibility Segregation (CQRS). Dependencies flow strictly inwards, ensuring that enterprise business rules and core domain logic remain completely decoupled from database frameworks, web presentation layers, and external microservice protocols.

Figure 4.2 depicts the logical layered architecture of the platform, detailing the functional components and dependency relationships across all tiers.



The logical system architecture comprises six cohesive tiers:

1. Presentation Tier (Angular Single Page Application):

- *User Interface Modules*: Encapsulates functional views for system administration, fleet master data, transit route planning, live cartographic tracking, and dynamic business intelligence dashboards.
- *GIS Cartographic Engine*: Built on `Leaflet.js` and `OpenStreetMap` tile layers, rendering dynamic vehicle markers, circuit polylines, geofenced collection perimeters, and live directional heading vectors.
- *Virtualized Data Grids*: Integrates `DevExtreme dx-data-grid` components to enable high-throughput tabular rendering, multi-column sorting, grouping, client-side filtering, and instant Excel/CSV data exports.
- *Reactive State and Interceptors*: Utilizes `RxJS` streams and custom HTTP interceptors to manage asynchronous event handling, automatic JWT Bearer token attachment, centralized error notifications, and global activity spinners.

2. API Gateway and Routing Tier (.NET 8 Web API):

- *Carter Minimal API Modules*: Employs modular endpoint slices (`ICarterModule`) that define RESTful routes with minimal memory footprint and zero controller boilerplate.
- *Authentication and Authorization*: Validates cryptographically signed JWT security tokens on incoming requests and evaluates dynamic navigation permissions via custom authorization handlers (`NavigationPermissionHandler`).
- *Global Exception Filter*: Intercepts application and domain exceptions, standardizing error responses into RFC 7807 Problem Details payloads with appropriate HTTP status codes.

3. Application and Use-Case Orchestration Tier (.NET 8 Application):

- *MediatR CQRS Pipeline*: Decouples request dispatchers from business handlers, dividing operations into explicit *Commands* (state mutations) and *Queries* (read-only data retrieval).
- *Pipeline Behaviors*: Executes a cross-cutting behavior chain comprising `LoggingBehavior` (structured request tracing), `ValidationBehavior` (fail-fast input schema validation via `FluentValidation`), and `UnitOfWorkBehavior` (atomic database commits for successful commands).
- *Data Transfer Mappings*: Uses compile-time `Mapster` mapping configurations to transform internal domain aggregates into lightweight, typed DTOs.

4. Domain Core Tier (.NET 8 Domain):

- *Pure Domain Aggregates*: Defines core business entities (`Bus`, `Modem`, `Chauffeur`, `Circuit`, `PointCollecte`, `Employe`, `Shift`, `OrdreMission`, `Rattachement`, `Societe`, `Utilisateur`, `RoleUtilisateur`) with private property setters and factory methods enforcing domain consistency.
- *Strongly-Typed Identifiers*: Implements ULID (Universally Unique Lexicographically Sortable Identifier) value objects, combining chronological sorting capabilities with URL-safe 128-bit uniqueness.
- *Domain Policies and Invariants*: Encapsulates pure business rules including Haversine great-circle distance calculations for stop geofencing, 15-minute telemetry freshness sliding windows, and vehicle seating capacity restrictions.
- *Universal Audit Contract*: Standardizes entity auditability through `AuditableEntity`, capturing creation and modification timestamps alongside actor identities.

5. Infrastructure and Persistence Tier (.NET 8 Infrastructure):

- *Entity Framework Core ORM*: Implements `ApplicationDbContext` with automated assembly scanning, fluent entity mappings, and database schema migrations.
- *Repository and Unit of Work*: Provides a generic `RepositoryBase<T>` and specialized domain repositories abstracting transactional data queries and batch updates.
- *Security and Cryptography*: Manages BCrypt password hashing (`PasswordService`) and HMAC-SHA256 token issuance (`JwtTokenGenerator`).
- *Document Generation Engine*: Employs `Puppeteer Sharp` to orchestrate a headless Chromium browser instance for rendering high-fidelity PDF work orders and operational reports.
- *External AI Connector*: Provides a resilient `HttpClient` client invoking the predictive machine learning microservice with automated timeout handling and fallback triggers.

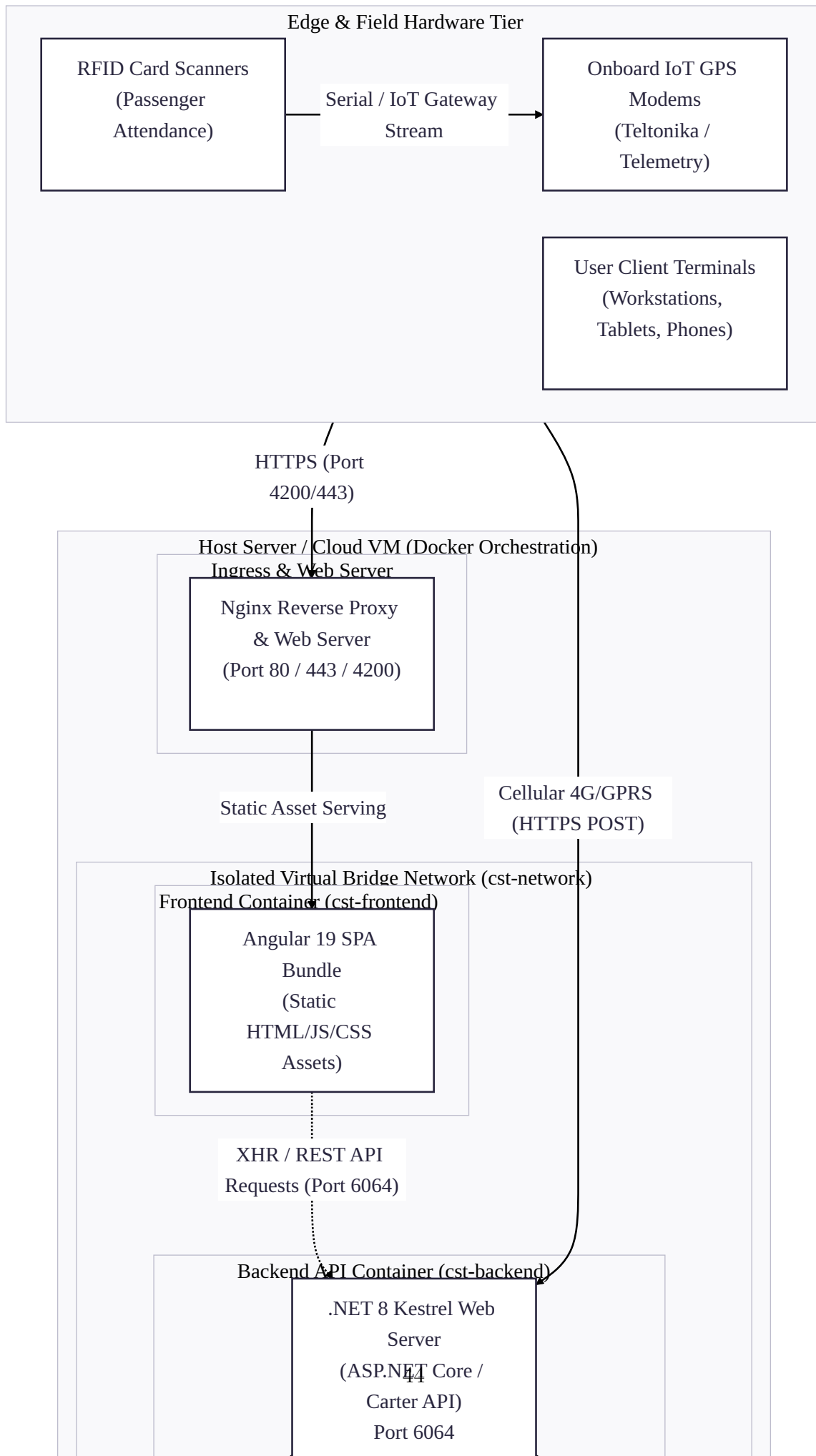
6. Predictive Intelligence Tier (Python FastAPI Microservice):

- *Asynchronous REST Endpoints*: Exposes high-performance asynchronous prediction routes (`/predict_eta`, `/batch_predict`) powered by Python 3.11 and the Uvicorn ASGI server.
- *Feature Preprocessing Engine*: Loads pre-trained scikit-learn standard scalers (`bus_eta_scaler.pkl`) to normalize spatial coordinates, speed vectors, temporal cyclical features, and occupancy ratios.
- *Predictive Regression Models*: Executes pre-trained regression models (`bus_eta_model.pkl`) to infer real-time arrival durations in minutes with sub-second inference latency.

4.2.3 Physical System Architecture

To guarantee seamless deployment, isolated runtime environments, and deterministic environment parity between local development, testing, and production servers, the entire CST LePoint ecosystem is containerized and orchestrated using **Docker** and **Docker Compose**.

Figure 4.3 illustrates the physical deployment topology, containerized service boundaries, virtual bridge networking, external port exposures, and persistent storage volume bindings.



The physical deployment infrastructure is structured across three hardware and containerized tiers:

1. Edge and Field Hardware Tier:

- *Vehicular IoT Modems*: Embedded telematics units (e.g., Teltonika devices) equipped with cellular SIM cards installed on buses. They capture GPS coordinates, velocity, and heading, transmitting HTTP/HTTPS POST payloads directly to the backend API ingestion endpoint over 4G/GPRS networks.
- *RFID Attendance Scanners*: Onboard badge scanners connected via serial or IoT gateway interfaces that register employee card taps during boarding and bundle them into real-time telemetry packets.
- *Client Terminals*: Desktop workstations, field tablets, and employee mobile devices executing standard web browsers to access the management portal over secure TLS connections (HTTPS).

2. Host Server and Container Orchestration Tier:

- *Virtual Bridge Network (`cst-network`)*: An isolated software-defined bridge network created by Docker Compose. Containers communicate internally using domain service names without exposing sensitive services (such as the database or ML engine) to the public internet.
- *Frontend Container (`cst-frontend`)*: Packages an Nginx 1.25 Alpine web server serving the optimized Angular production bundle. It listens on container port 80, mapped to host port 4200 for user web traffic.
- *Backend API Container (`cst-backend`)*: Packages the pre-compiled .NET 8 Web API runtime executing on the high-performance Kestrel web server on host port 6064. It executes automatic schema migrations at startup and orchestrates business operations.
- *Database Container (`cst-database`)*: Runs Microsoft SQL Server 2019 Standard on internal port 1433, enforcing transactional integrity, row-level locking, and multi-tenant partitioning.
- *AI Microservice Container (`cst-ml-service`)*: Runs a Python 3.11 environment with Uvicorn and FastAPI on internal port 8001, executing real-time machine learning inference routines.

3. Persistent Storage and Volume Tier:

- *Database Storage Volume (`cst_db_data`)*: A dedicated host volume binding SQL Server data files (`.mdf`), transaction log files (`.ldf`), and indexes, ensuring absolute

data durability across container restarts, migrations, or host reboots.

- *Model Artifacts Volume* (`cst_ml_models`): A persistent directory volume mounting serialized machine learning models (`.pkl` files), enabling seamless model retraining, hot-swapping, and version updates without requiring container rebuilds.

Table 4.1 summarizes the physical deployment specifications, network port mappings, and communication protocols across all containerized services.

Container Service	Base Image / Runtime	Port (Host:Cont)	Protocol
cst-frontend	nginx:1.25-alpine	4200:80	HTTP / HTTPS
cst-backend	mcr.microsoft.com/dotnet/aspnet:8.0	6064:6064	REST / HTTPS
cst-database	mcr.microsoft.com/mssql/server:2019	1433:1433	TCP (TDS)
cst-ml-service	python:3.11-slim	8001:8001	HTTP (JSON)

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Table 4.1: Containerized Services Deployment Specification and Port Mappings

4.2.4 End-to-End Operational Lifecycle

The functional strength of the platform lies in unifying the entire lifecycle of enterprise transit into an automated, continuous operational pipeline.

Figure 4.4 outlines the five distinct operational stages that govern daily transit workflows.

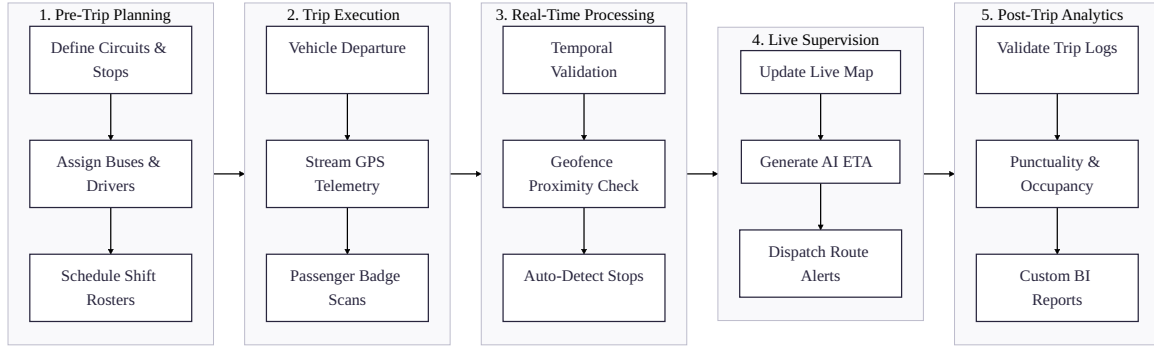


Figure 4.4: End-to-End Operational Transit Lifecycle

This lifecycle unfolds across five consecutive phases:

1. **Pre-Trip Planning:** Operational coordinators define circuits, map pickup stations, assign dedicated vehicles and qualified drivers, and match employee shifts to designated bus routes.
2. **Trip Execution:** As the vehicle commences its scheduled circuit, onboard IoT devices stream real-time GPS coordinates and register passenger RFID badge taps during boarding.
3. **Real-Time Processing:** Ingested data packets undergo immediate temporal verification, spatial proximity calculation against scheduled geofences, and automated stop identification.
4. **Live Supervision:** Dispatchers monitor fleet progress on an interactive cartographic map, while passengers consult dynamically updated estimated arrival times and receive notifications for deviations.
5. **Post-Trip Analytics:** Upon trip completion, operational data is aggregated to compute key performance metrics, vehicle occupancy rates, driver punctuality, and exportable financial reconciliation reports.

4.2.5 Inter-Tier Communication and Fault-Tolerance Protocols

To maintain operational continuity across fluctuating cellular connectivity and high concurrency spikes, the architecture implements robust communication and resilience protocols:

- **Stateless REST and Cryptographic Bearer Tokens:** All communication between the Angular web client and the backend API occurs over stateless HTTPS RESTful requests carrying signed JSON Web Tokens (JWT). User identity, tenant context (`SocieteId`), and authorization privileges are cryptographically validated per request without server-side session overhead.
- **Sliding-Window Telemetry Replay Defense:** Incoming telemetry packets from IoT modems are validated against a sliding temporal window of 15 minutes. Stale packets resulting from cellular network retransmissions are acknowledged but flagged, preventing historical packet replays from corrupting live cartographic state.
- **Predictive Inference Timeout and Resilient Fallback:** When invoking the machine learning microservice for ETA forecasts, the backend enforces an asynchronous 2-second timeout threshold. If the AI service experiences latency or temporary unavailability, the system triggers a graceful fallback calculating arrival duration via a deterministic Haversine distance heuristic ($v_{\text{avg}} = 40 \text{ km/h}$). This guarantees that user interfaces always display actionable transit timing without hanging.
- **Transactional Unit-of-Work Consistency:** State mutations involving multiple relational entities (e.g., dispatching work orders, assigning employee attachments, or logging geofenced stop arrivals) execute within atomic database transactions via the Unit of Work pattern, ensuring full ACID consistency and zero partial data corruption.

4.3 Multi-Tenant Identity and Access Governance

4.3.1 Multi-Organization Partitioning

Enterprise transit management frequently involves multiple subsidiaries, partner companies, or client organizations operating under a shared software infrastructure. To guarantee strict data privacy, operational independence, and administrative isolation, the platform incorporates a shared-schema multi-tenancy model.

Every operational entity—including vehicles, IoT modems, employee records, transit circuits, work orders, and historical telemetry logs—is intrinsically scoped to an organization

identifier. This structural partition guarantees that:

- Users belonging to a specific company are restricted strictly to their organization's assets and transit schedules;
- Corporate managers have full autonomous control over their fleet, workforce rosters, and role assignments without risking cross-tenant data leaks;
- Central administrators maintain cross-organization visibility for global system maintenance and infrastructure health monitoring.

4.3.2 Dynamic Role-Based Access Control

To accommodate the diverse hierarchy of transit personnel, the system implements a dynamic Role-Based Access Control (RBAC) architecture. Rather than relying on hardcoded permissions, the system models permissions as a dynamic matrix mapping functional navigation modules to granular action capabilities: *Read*, *Add*, *Edit*, and *Delete*.

Figure 4.5 illustrates the access governance workflow from initial request to dynamic UI presentation.

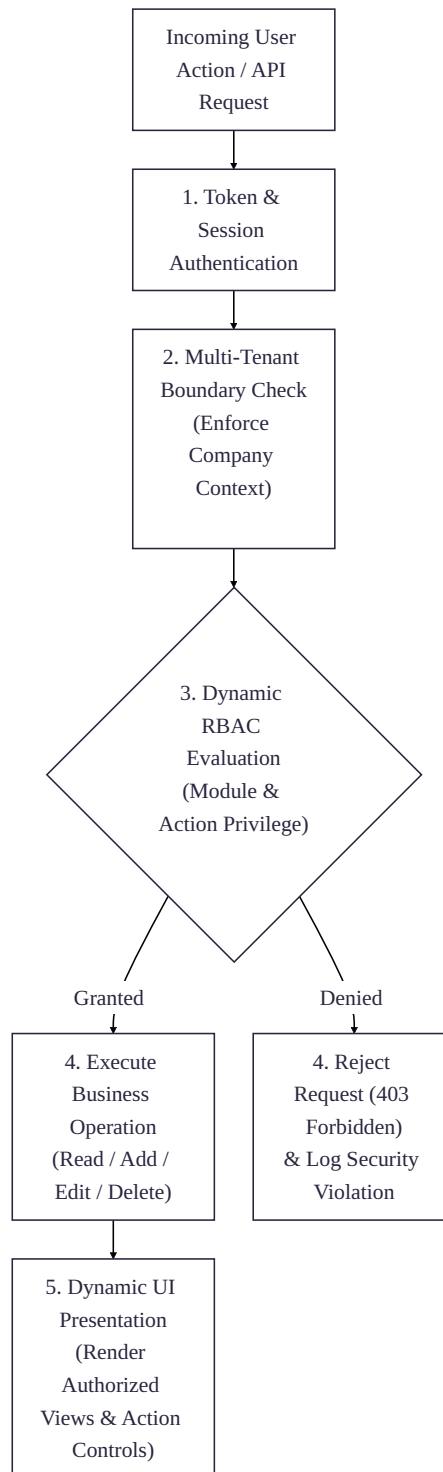


Figure 4.5: Multi-Tenant & Dynamic RBAC Governance Flow

The access governance mechanism enforces security at both user interface and backend tiers:

- **Dynamic Interface Filtering:** Upon user sign-in, the client interface inspects the

serialized permission matrix associated with the user's role. Navigation menus, action buttons, and route views are dynamically filtered to display only authorized features.

- **Inviolable Backend Validation:** Every operational request is independently verified at the application boundary. If an incoming command requests an unauthorized action or accesses resources outside the authenticated company context, the request is terminated immediately with an HTTP 403 Forbidden response.

4.3.3 Secure Session and Authentication Management

Authentication is established through a stateless token-based architecture:

- **Credential Verification:** User passwords undergo adaptive cryptographic hashing, preventing plain-text exposure and defending against dictionary attacks.
- **Stateless Signed Tokens:** Upon successful authentication, the system generates a cryptographically signed security token containing user claims (user identity, organizational context, and assigned role permissions).
- **Seamless Session Restoration:** When a user opens or refreshes the application, a lightweight validation probe verifies token validity and retrieves the latest role permissions, restoring the user session smoothly without requiring repeated credential inputs.

4.4 Fleet and Transit Logistics Planning Subsystem

4.4.1 Fleet Assets and IoT Unit Management

The foundation of effective transit logistics rests upon maintaining a synchronized catalog of physical and operational assets:

- **Vehicular Fleet Management:** Captures vehicle profiles, seating capacities, registration identifiers, and operational statuses (active, in-maintenance, decommissioned).
- **Hardware Modem Pairing:** Manages onboard IoT units by pairing unique hardware identifiers (such as 15-digit International Mobile Equipment Identity numbers) to specific vehicles, ensuring that all incoming telemetry is mapped to the correct bus.
- **Driver Registry and Qualifications:** Records driver profiles, contact details, assigned companies, and driving credentials to ensure that only authorized personnel are scheduled for transit execution.

4.4.2 Circuit and Geolocation Pickup Route Planning

Circuit design models the spatial and temporal paths that vehicles follow during daily transit:

- **Circuit Modeling:** Defines named transit routes, departure origins, destination sites, estimated roundtrip durations, and operating schedules.
- **Pickup Station Mapping:** Each circuit is composed of an ordered sequence of designated pickup stops. Every stop is defined by geographical coordinates (latitude and longitude) and assigned a configurable geofence radius (typically 250 meters).
- **Route Sequence Enforcement:** Ordered sequence indexes ensure that transit progress and arrival countdowns can be computed deterministically as the vehicle navigates from the first stop to the destination terminus.

4.4.3 Workforce Shift and Bus Roster Scheduling

To ensure balanced vehicle capacity and punctuality, the platform coordinates workforce transit rosters:

- **Shift Alignment:** Work shifts (e.g., morning shift, afternoon shift, night shift) define standard working hours and operational transit windows.
- **Employee Pickup Assignment:** Employees are assigned to a designated circuit and a specific pickup station based on geographical proximity to their residence.
- **Work Order Execution Plans:** Dispatchers generate work orders that bind a designated bus, an assigned driver, and a scheduled circuit to a specific operating date and shift, creating an actionable schedule for daily execution.

4.4.4 Conceptual Domain Information Model

Figure 4.6 depicts the unified conceptual domain information model, illustrating the core business entities and their relational dependencies across the entire transit platform.

This conceptual information model highlights the key structural relationships:

- **Organization Root:** The `Societe` entity acts as the organizational boundary for all users, roles, fleet assets, circuits, employees, and shifts.
- **Operational Synthesis (`OrdreMission`):** Unifies the operational execution triangle—combining a specific vehicle, driver, circuit, and shift for a designated calendar date.

- **Workforce Attachment (`RattachementEmploye`):** Links individual employees to their designated pickup stops and transit circuits.
- **Telemetry and Event Tracking:** Captures real-time GPS locations (`PositionBus`) and passenger attendance scans (`PointageEvenement`) referencing both vehicles and stops.

4.5 Real-Time Telemetry Processing and Automated Supervision

4.5.1 High-Throughput Telemetry Ingestion Pipeline

The telemetry processing engine is engineered to ingest, validate, and persist high-frequency data packets transmitted by vehicle modems across the entire fleet:

- **Data Ingestion Protocol:** Onboard modems periodically transmit HTTP POST packets containing hardware identifiers, geographical coordinates, passenger occupancy counters, and UTC timestamps.
- **Temporal Skew Protection:** To prevent replay attacks and reject stale data caused by transient cellular connectivity drops, incoming packets are checked against a sliding temporal window (rejecting data with timestamp deviations exceeding 15 minutes).
- **Modem-to-Fleet Resolution:** The system resolves the hardware identifier to its corresponding active vehicle and retrieves the circuit associated with the active work order.

4.5.2 Live Cartographic Supervision

The supervision portal provides fleet dispatchers and operations managers with an interactive, live-updating cartographic command center:

- **Dynamic Map Rendering:** Renders geographical base maps overlaid with active vehicle markers, color-coded circuit polylines, pickup stations, and company destination sites.
- **Vehicle State Visualization:** Bus markers visually reflect operational status (in transit, idling, stopped, out of radius) and display real-time passenger capacity badges.
- **Continuous Position Synchronization:** The client interface receives streamed

position updates, smoothly repositioning vehicle markers on the map without full page reloads.

4.5.3 Automated Geofencing and Stop Detection Engine

A core innovation of the platform is the elimination of manual stop recording through automated spatial geofencing.

Figure 4.7 illustrates the sequence execution flow from telemetry transmission to automated geofence evaluation and map broadcasting.

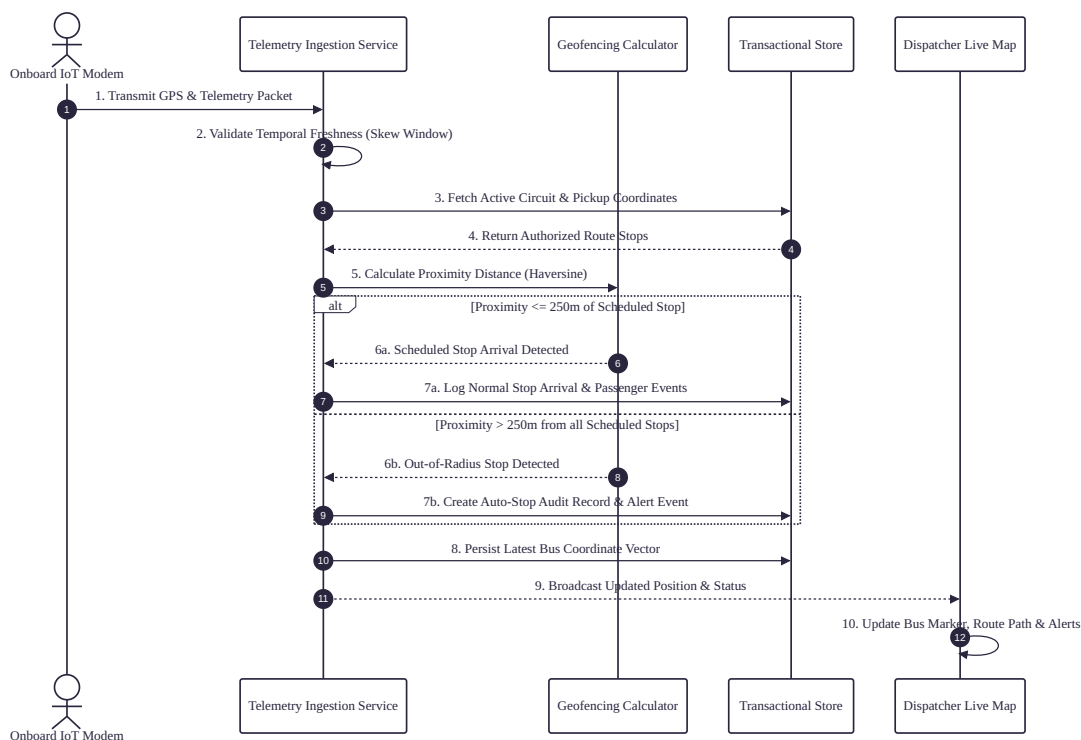


Figure 4.7: Real-Time Telemetry Ingestion & Automated Geofence Detection Pipeline

The automated geofencing logic proceeds as follows:

1. **Haversine Distance Calculation:** When a telemetry packet is received, the engine computes the great-circle distance between the vehicle's current coordinates and all pickup stops assigned to its active circuit.
2. **Normal Stop Detection:** If the bus is within the authorized 250-meter radius of a scheduled stop, the system logs a verified stop arrival and associates any concurrent passenger badge scans with that station.

3. **Unscheduled Stop Identification:** If passenger boarding scans or sustained stops occur while the vehicle is located more than 250 meters away from all scheduled stations, the engine automatically creates an audit record (e.g., PC-AUTO) and flags the event as an out-of-radius deviation on the dispatcher’s live map.

4.6 Predictive Intelligence and Transit Forecasting

4.6.1 Real-Time Estimated Time of Arrival (ETA) Forecasting

Uncertainty regarding vehicle arrival times is a major source of employee dissatisfaction and productivity loss in corporate transportation. To address this challenge, the platform incorporates a dedicated machine learning microservice that calculates real-time Estimated Time of Arrival (ETA) predictions for every upcoming pickup stop.

Figure 4.8 depicts the end-to-end forecasting pipeline connecting the client portal, business orchestrator, and predictive machine learning engine.

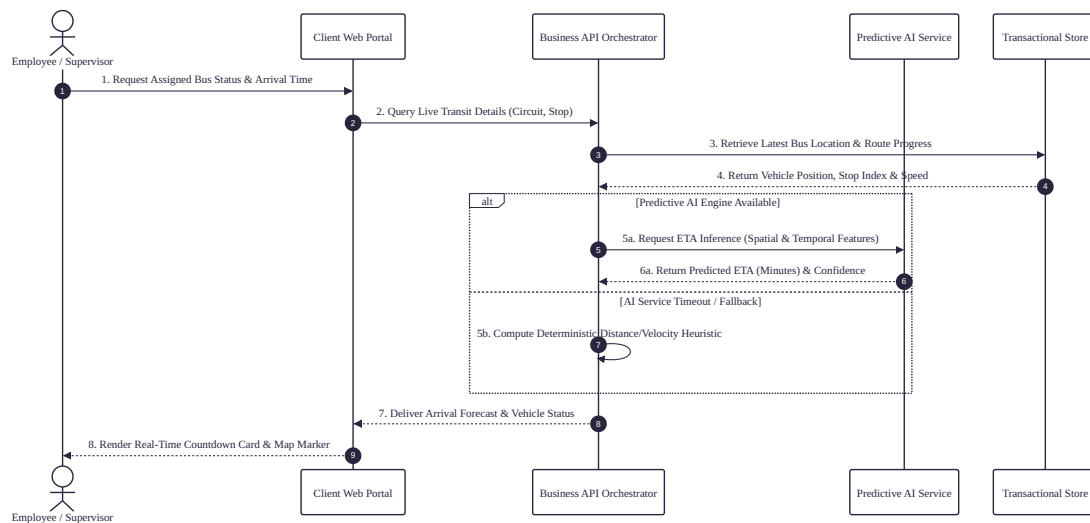


Figure 4.8: Predictive ETA Forecasting & Supervision Intelligence Flow

The forecasting engine evaluates a rich set of operational and environmental features:

- **Spatial Metrics:** Remaining great-circle distance to the target pickup station, remaining number of intermediate stops, and current vehicle transit speed;
- **Temporal Dynamics:** Hour of the day, cyclical time encodings (sine and cosine transforms to capture continuous daily cycles), day of the week, and rush-hour indicator flags;

- **Transit Variables:** Current bus passenger occupancy ratio and historical route completion durations.

4.6.2 Passenger and Driver Roster Consultation

The predicted arrival values directly feed user-facing interfaces:

- **Employee Commute Card:** Employees can view their assigned bus route, current vehicle location on the map, and a live countdown showing the expected arrival time at their personal pickup stop.
- **Driver Schedule View:** Drivers receive a sequence of upcoming stops with target arrival times, assisting them in maintaining optimal transit pacing.

4.6.3 Fault-Tolerant Fallback Strategy

To guarantee uninterrupted service availability during network degradation or AI service maintenance, the platform implements a dual-tier resilience model:

- If the machine learning microservice does not respond within a configured timeout window (e.g., 2 seconds), the backend automatically falls back to a deterministic calculation based on spatial route distance and historical average velocity (40 km/h).
- The client interface seamlessly displays the estimated duration alongside a subtle indicator, ensuring that supervisors and passengers always receive actionable arrival guidance without encountering application errors.

4.7 Operational Analytics and Business Intelligence Subsystem

4.7.1 Operational Key Performance Indicators (KPIs)

To enable continuous process improvement and data-driven decision-making, the platform aggregates historical transit and attendance logs into key operational indicators:

- **Fleet Punctuality and Schedule Adherence:** Evaluates planned versus actual arrival times across all pickup stops and terminal destinations, highlighting chronic bottleneck routes.

- **Vehicle Occupancy and Capacity Utilization:** Analyzes passenger boarding counts against vehicle seating capacities, identifying under-utilized buses or overcrowded circuits requiring fleet rebalancing.
- **Attendance and Boarding Compliance:** Cross-references scheduled employee shift rosters with recorded RFID badge scans, summarizing presence rates, unboarded passengers, and unauthorized pickups.
- **Operational Route Costs:** Computes total distance traveled, transit durations, and vehicle operational hours for logistics cost optimization.

4.7.2 Dynamic Reporting Perspectives and Layout Customization

To accommodate the diverse analytical needs of executive directors, HR managers, and logistics dispatchers, the Business Intelligence (BI) subsystem provides dynamic data exploration tools:

- **Interactive Tabular Exploration:** Users can perform multi-level grouping (e.g., grouping by company, then by shift, then by circuit), multi-column sorting, and multi-criteria date filtering across massive operational datasets.
- **Layout State Persistence:** Analysts can customize their data grid view—arranging column orderings, applying custom filters, and defining visibility preferences—and save these configurations as named personal or organizational layouts. Saved layouts are persisted in the database and restored automatically upon subsequent visits.

4.7.3 Data Export and Logistics Reconciliation

To integrate transit intelligence into broader enterprise workflows, the analytics subsystem provides robust data export capabilities:

- **Multi-Format Reports:** Operational summaries and detailed attendance records can be exported directly to standard document formats including PDF, Excel, and CSV.
- **Billing and Provider Reconciliation:** Exported reports serve as authoritative audit logs for validating third-party transport provider invoices against actual trips and kilometers executed.

4.8 Architectural Cohesion and System Integrity

4.8.1 Separation of Concerns and Modular Design

To guarantee that the platform remains maintainable, extensible, and robust over its lifecycle, the system is designed around strict separation of architectural concerns:

- **Decoupled Domain Core:** Core business logic, validation rules, and domain policies are completely isolated from user interface libraries, external frameworks, and database persistence mechanisms.
- **Segregation of Read and Write Operations:** By separating write-heavy transactional operations (such as high-throughput telemetry ingestion and vehicle updates) from read-heavy analytical queries (such as live map streaming and BI reporting), the system prevents database contention and maintains high responsiveness under heavy operational loads.

4.8.2 Data Consistency and Transactional Auditability

Enterprise transit data demands absolute consistency and complete traceability:

- **Transactional Unit-of-Work:** All modifications involving multiple related entities (such as updating a work order and reassigning a driver) are executed within atomic transactional boundaries, guaranteeing that operations either succeed entirely or roll back safely.
- **Universal Audit Trail:** Every domain entity inherits standardized audit tracking (recording creation timestamps, creator identities, modification timestamps, and modifier identities). This provides an immutable operational audit log for administrative oversight and security compliance.

4.9 Conclusion

This chapter established the comprehensive architectural and conceptual design of the CST LePoint platform. By architecting a multi-tier distributed ecosystem, the proposed solution directly addresses the operational bottlenecks identified in the legacy system analysis:

- Multi-tenant identity governance and dynamic role permissions ensure secure, segregated enterprise access;
- Integrated logistics planning structures fleet assets, circuits, and workforce rosters;

- High-throughput telemetry ingestion and automated geofencing eliminate manual stop logging and provide live cartographic fleet visibility;
- Predictive machine learning microservices deliver accurate real-time arrival forecasts;
- The Business Intelligence subsystem transforms raw operational logs into actionable KPIs and customizable analytics.

With the functional architecture and conceptual models defined, Chapter 5 transitions from abstract design to concrete technical realization, detailing the development stacks, client-side modules, server-side handlers, container deployment topologies, and empirical performance evaluations.

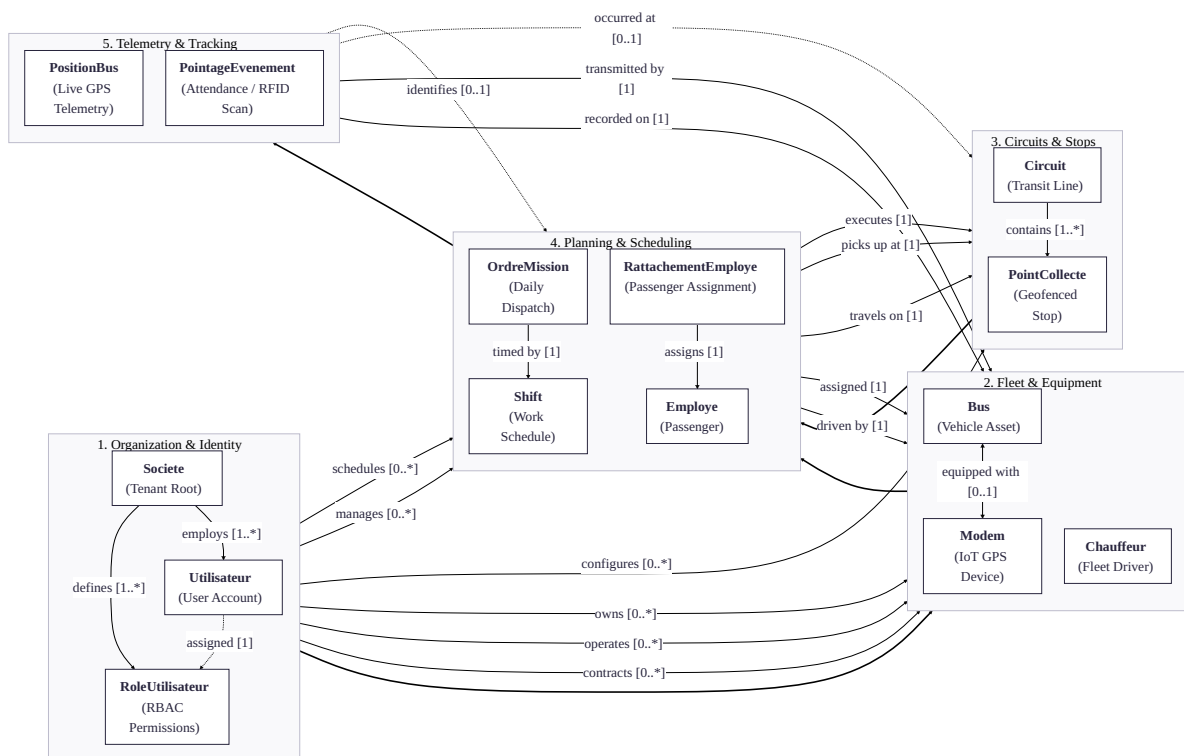


Figure 4.6: Unified Conceptual Domain Information Model

5 Implementation and Realization

5.1 Introduction

Following the conceptual design and architectural blueprints established in Chapter 4, this chapter presents the concrete implementation, technological realization, and empirical evaluation of the CST LePoint platform. It bridges abstract models with production-grade engineering artifacts across the entire software development lifecycle.

In this chapter, we systematically detail:

- The technology stack selection across backend, frontend, database, AI microservice, and DevOps containerization tiers;
- The step-by-step implementation of functional modules, featuring user interface realizations, sequence execution flows, endpoint logic, C# handlers, validation rules, and Python machine learning inference;
- Automated testing strategies, system ingestion latency benchmarks, and a comprehensive Requirements Traceability Matrix validating full coverage of Chapter 3.

5.2 Technological Environment and Stack Selection

To fulfill the performance, maintainability, and data integrity requirements of enterprise transit supervision, the system is engineered using modern open-source frameworks, industry-standard development tooling, and containerized deployment infrastructure.

5.2.1 Core Application Frameworks and Persistence

Figure 5.1 illustrates the primary application runtime engines and persistence platforms powering the system tiers.



Figure 5.1: Core Technology Stack and Application Frameworks

The responsibilities of the core application tiers are organized as follows:

- **Backend Web API (.NET 8 C#):** Structured according to Clean Architecture and CQRS principles. Minimal APIs (**Carter**) deliver high-throughput, low-latency endpoint routing, **MediatR** orchestrates in-process command and query handlers, **FluentValidation** enforces strict fail-fast payload verification, and **Entity Framework Core 8** manages object-relational mapping and database migrations.
- **Client Web Interface (Angular 19):** A responsive Single Page Application (SPA) utilizing **RxJS** for reactive event stream handling, **Leaflet.js** for dynamic cartographic supervision and live vehicle positioning, **DevExtreme** for high-performance virtualized datagrids, and **ngx-translate** for multi-language dictionary resolution.
- **Predictive Intelligence Service (Python FastAPI):** An asynchronous microservice running Python 3.11, packaging pre-trained machine learning regression engines (**Scikit-Learn** and **LightGBM**) for real-time arrival forecasting and spatial Great-Circle Haversine distance calculations.
- **Relational Database (Microsoft SQL Server 2019):** Enterprise ACID-compliant transactional data store ensuring relational integrity, multi-tenant organization partitioning, audit trail persistence, and operational log management.

5.2.2 DevOps, Cloud, and API Documentation Infrastructure

Figure 5.2 shows the infrastructure, containerization, and API contract platforms utilized for deployment and testing.



Figure 5.2: DevOps, Cloud Infrastructure, and API Documentation Platforms

- **Container Orchestration (Docker & Docker Compose):** Packages the Angular

frontend (Nginx), .NET 8 WebAPI, SQL Server database, and Python FastAPI microservice into isolated container environments communicating over a private internal bridge network.

- **Cloud Infrastructure (Microsoft Azure):** Provides enterprise cloud hosting capabilities, scalable compute infrastructure, and reliable managed deployment environments.
- **API Specification and Contract Discovery (Swagger / OpenAPI):** Delivers interactive, self-documenting API endpoints, schema definitions, and testing interfaces for backend RESTful routes.

5.2.3 Development Environments and Version Control Tooling

Figure 5.3 displays the integrated development environments (IDEs) and source control platforms used throughout engineering.



Figure 5.3: Integrated Development Environments and Collaboration Tooling

- **Backend Development IDE (JetBrains Rider):** Cross-platform .NET IDE providing rich C# static code analysis, deep refactoring capabilities, Unit-of-Work debugging, and integrated database querying.
- **Frontend Development IDE (JetBrains WebStorm):** Specialized TypeScript and Angular IDE offering real-time component syntax inspection, template validation, and reactive debugging.
- **Source Control and Collaboration (GitHub):** Distributed version control system hosting the codebase, managing branch lifecycles, tracking issues, and facilitating collaborative code reviews.

5.3 Module Implementations

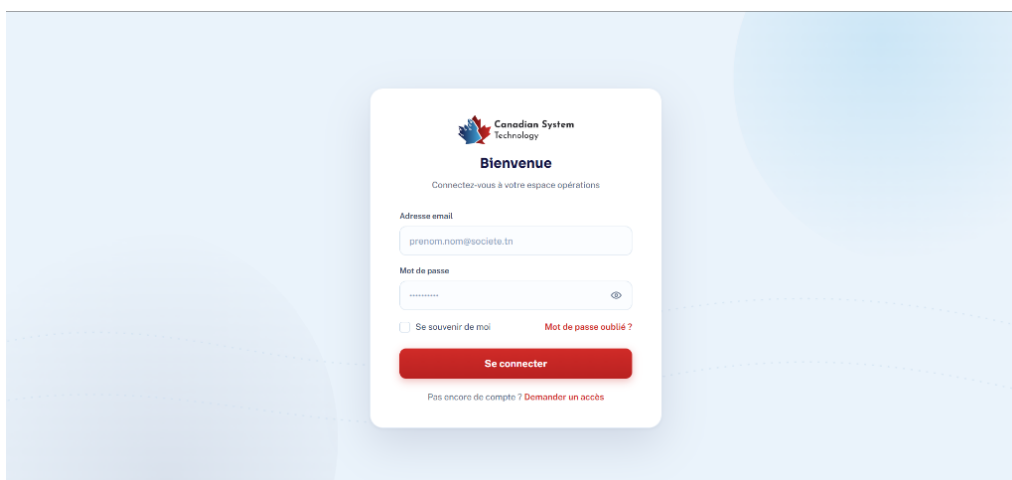
This section presents the concrete realization of the platform's core functional modules, illustrating user interface screens, interaction workflows, and backend execution logic.

5.3.1 Module 1: Authentication and Access Governance

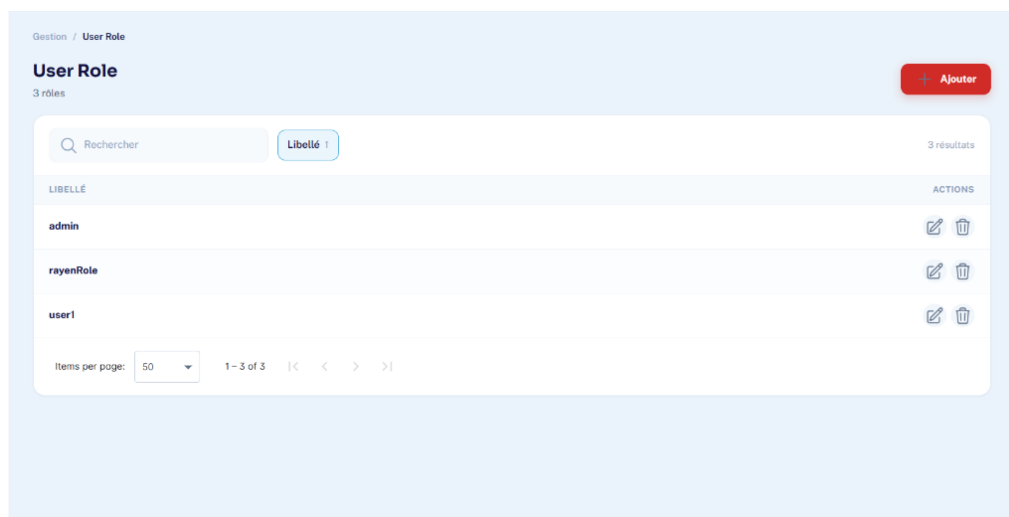
The authentication module delivers multi-tenant identity verification, password hashing, stateless JSON Web Token (JWT) issuance, and dynamic Role-Based Access Control (RBAC).

User Interface Realization

Figure 5.4 presents the user interface screens for user authentication and role-based privilege administration.



(a) Multi-Tenant Authentication and Sign-In Screen



(b) Role-Based Access Control and Permission Governance Interface

Figure 5.4: User Authentication and Role Management Interfaces

As shown in Figure 5.4a, the authentication interface features a minimalist, card-based login dialogue customized with corporate branding. It captures the user's email address

and password, providing features such as password visibility toggling, session persistence (*Remember Me*), and password recovery triggers. Client-side Angular reactive validators enforce format constraints prior to dispatching network requests.

Figure 5.4b depicts the user role administration screen within the governance sub-system. Administrators can search existing roles, define custom privilege groups (e.g. `admin`, `user1`), and assign granular module authorizations spanning Read, Add, Edit, and Delete actions across application routes.

Sequence Execution Flow

Figure 5.5 illustrates the end-to-end token issuance and verification pipeline executed upon user sign-in.

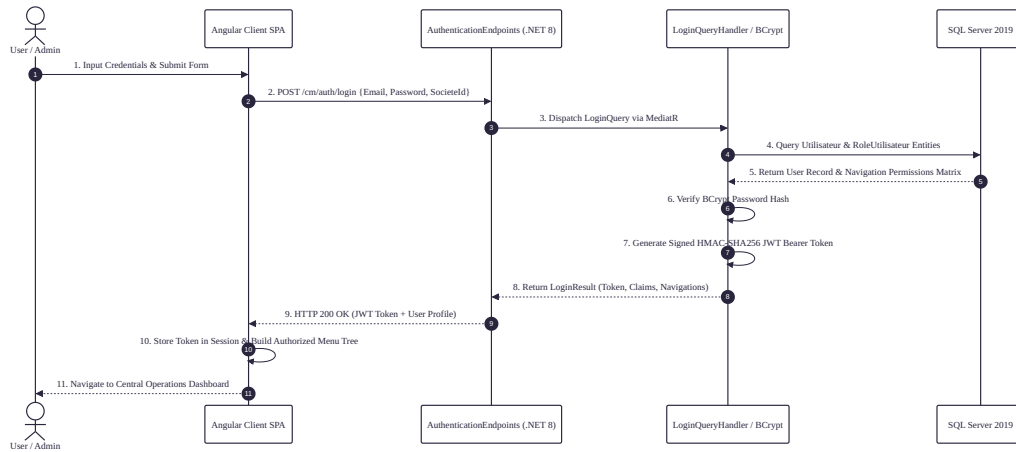


Figure 5.5: Authentication and Security Token Issuance Flow

Technical Implementation Highlights

- **BCrypt Password Verification:** Passwords submitted during login are evaluated against stored cryptographic hashes using adaptive BCrypt work factors.
- **Stateless JWT Generation:** Upon successful verification, the backend generates an HMAC-SHA256 signed JWT containing essential claims (`UserId`, `SocieteId`, `RoleUtilisateurId`) and the serialized navigation permissions matrix.
- **Angular Navigation Guards:** On the client side, Angular functional route guards inspect token validity and decoded role permissions, automatically hiding unauthorized navigation links and blocking forbidden route transitions.

5.3.2 Module 2: Operations Management and Master Data Logistics

The master data module forms the operational backbone of the platform, managing executive key performance indicators, vehicle fleets, drivers, transit circuits, geocoded pickup stops, workforce directories, and shift schedules.

Operational Overview and Central Dashboard

Figure 5.6 displays the central operations dashboard accessible to dispatchers and operational managers upon authentication.

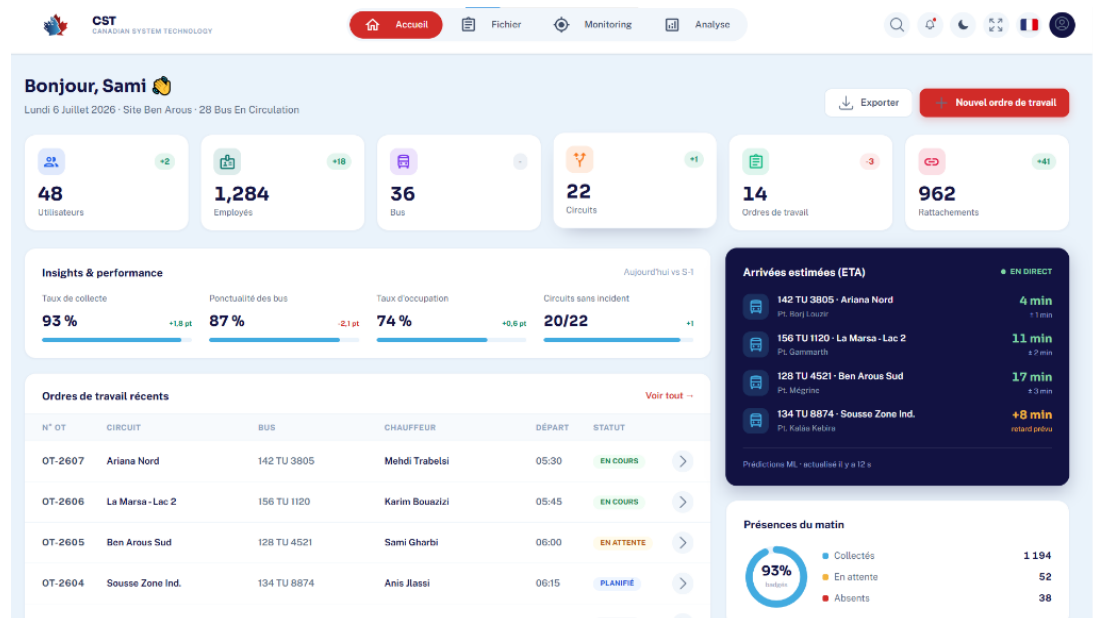


Figure 5.6: Central Operations and Real-Time Telemetry Overview Dashboard

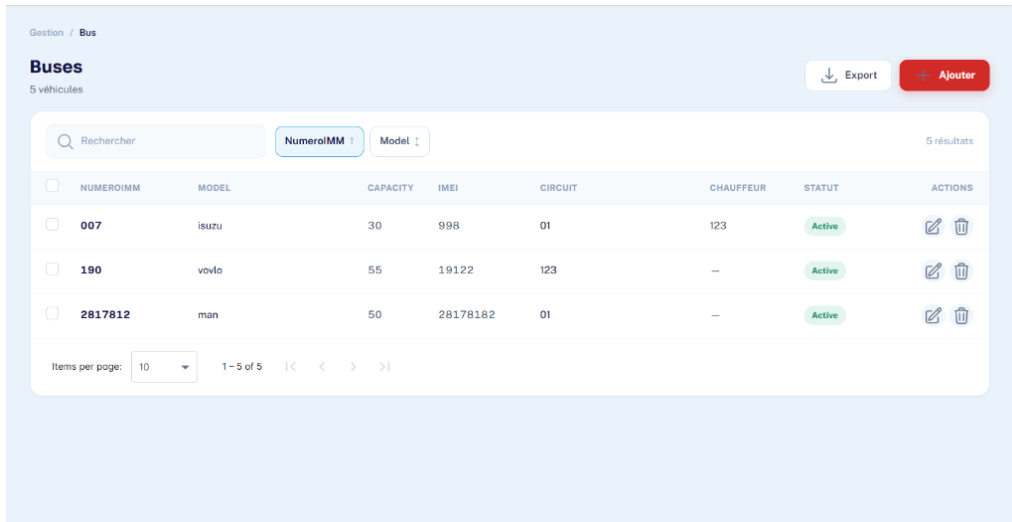
The central operations dashboard aggregates critical operational metrics into an executive cockpit:

- **Operational KPI Summary Cards:** Displays real-time counts for active system entities, including total Users (48), Registered Employees (1,284), Active Buses (36), Defined Circuits (22), Scheduled Work Orders (14), and Employee Route Attachments (962).
- **Insights and Performance Gauges:** Tracks real-time service quality indicators, including Collection Rate (93%), Bus Punctuality (87%), Average Seating Occupancy (74%), and Incident-Free Circuits (20/22).

- **Recent Work Orders (*Ordres de Travail*):** A live table tracking current transport work orders with real-time lifecycle states (EN COURS, EN ATTENTE, PLANIFIÉ, TERMINÉ), departing timestamps, and driver-to-bus allocations.
- **Live Machine Learning Arrival Forecasts (ETA):** An embedded live ticker computing real-time remaining arrival minutes (e.g. *142 TU 3805 – Ariana Nord: 4 min $\pm$ 1 min*) dynamically refreshed via background polling.
- **Morning Attendance Distribution:** A circular donut chart summarizing daily boarding execution (93% badged, 1,194 scanned passengers, 52 pending, 38 absentees).

Fleet and Driver Logistics Management

Figure 5.7 illustrates the master data catalogs for managing transit vehicles and licensed drivers.



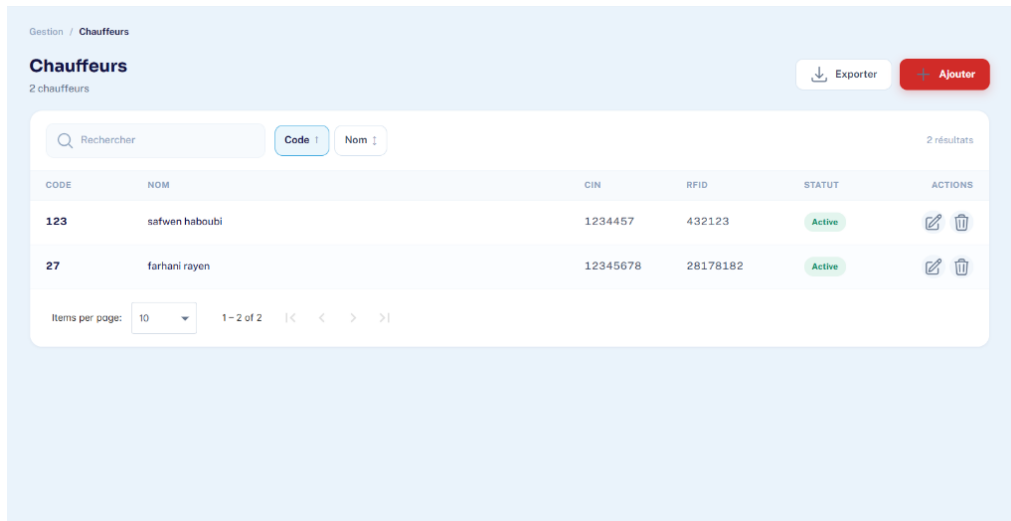
Buses
5 véhicules

Rechercher **NumeroIMM** **Model** 5 résultats

	NUMEROIMM	MODEL	CAPACITY	IMEI	CIRCUIT	CHAUFFEUR	STATUT	ACTIONS
☐	007	isuzu	30	998	01	123	Active	
☐	190	volvo	55	19122	123	—	Active	
☐	2817812	man	50	28178182	01	—	Active	

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(a) Fleet Vehicle Catalog and IoT Modem IMEI Pairing



Chauffeurs
2 chauffeurs

Rechercher **Code** **Nom** 2 résultats

	CODE	NOM	CIN	RFID	STATUT	ACTIONS
	123	safwen haboubi	1234457	432123	Active	
	27	farhani rayen	12345678	28178182	Active	

Items per page: 10 1 - 2 of 2 < >

(b) Driver Directory and RFID Card Association

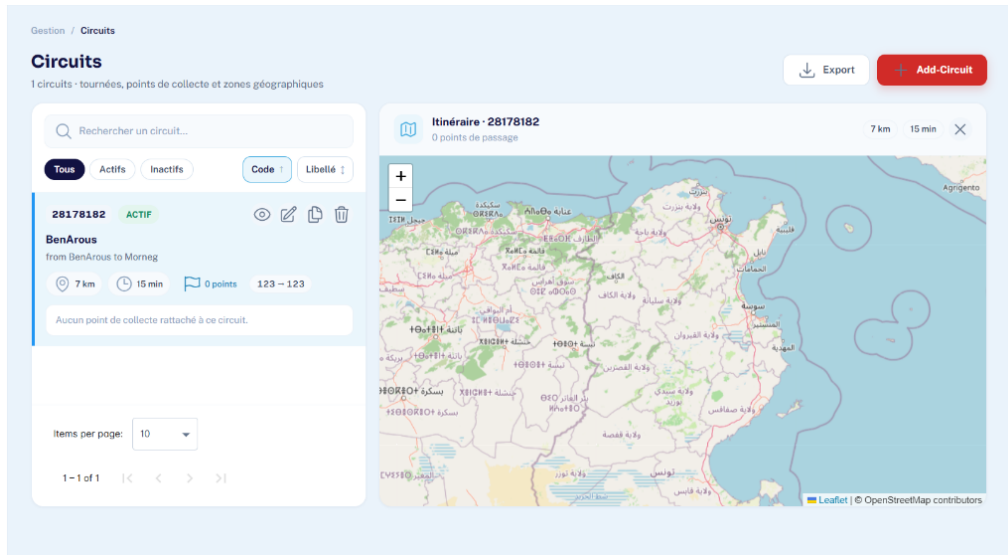
Figure 5.7: Fleet Vehicle and Driver Management Interfaces

As shown in Figure 5.7a, the bus management interface catalogs all company transit vehicles. Each record tracks the vehicle registration plate (**NumeroIMM**), bus model, seating capacity, assigned driver, operational status (**Active/Inactive**), and the 15-digit hardware IMEI of the onboard IoT telemetry modem.

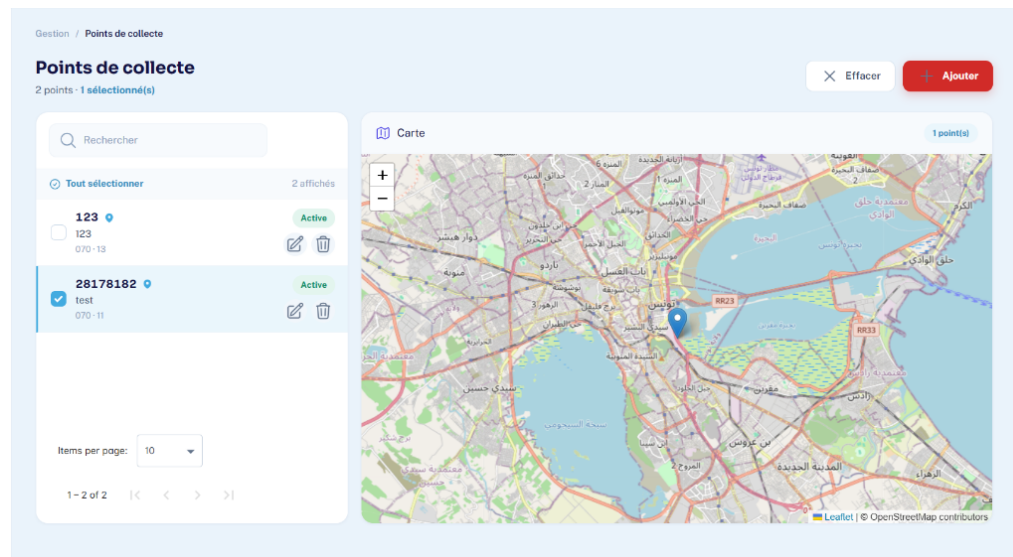
Figure 5.7b presents the driver management interface. Drivers are identified by unique driver codes, national identity card numbers (**CIN**), and dedicated RFID card numbers used for driver authentication upon boarding transit vehicles.

Transit Circuits and Geocoded Pickup Points

Figure 5.8 displays the interactive route planning and pickup point geocoding interfaces.



(a) Transit Circuit Planning with Split-View Cartographic Display



(b) Geocoded Pickup Points Catalog with OpenStreetMap Spatial Pins

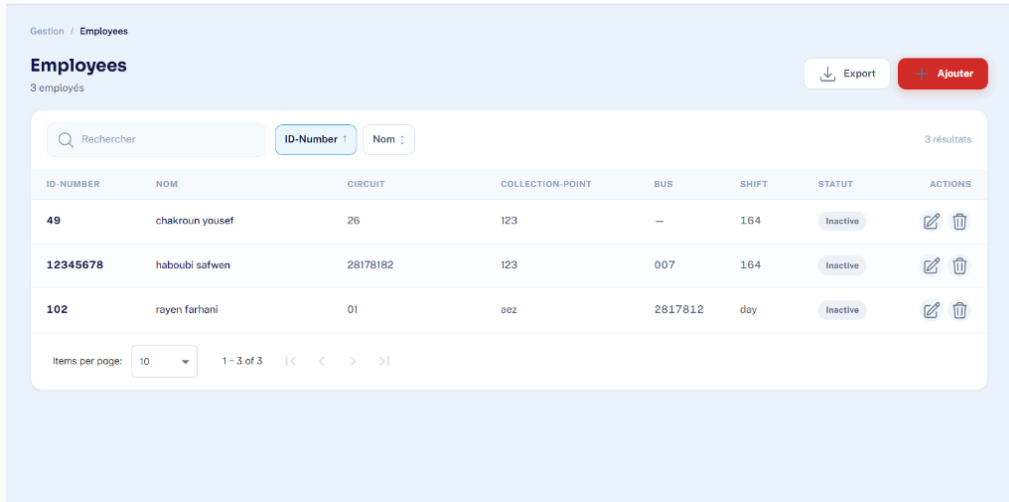
Figure 5.8: Transit Circuit Planning and Pickup Geocoding Interfaces

Figure 5.8a demonstrates the split-screen circuit management interface. The left panel lists defined transit lines with estimated distance in kilometers, projected journey duration in minutes, and assigned pickup stations. The right panel leverages `Leaflet.js` and `OpenStreetMap` to project the geographic route path across the Tunisian territory.

Figure 5.8b shows the pickup point management interface. Dispatchers can define named collection stations, associate them with specific geographic coordinates, and visually verify stop placement on the interactive map before binding them to transit circuits.

Workforce Directory, Shifts, and Roster Assignments

Figures 5.9 and 5.10 showcase the employee directory, shift configuration, and employee-to-pickup attachment interfaces.



Gestion / Employees

Employees
3 employés

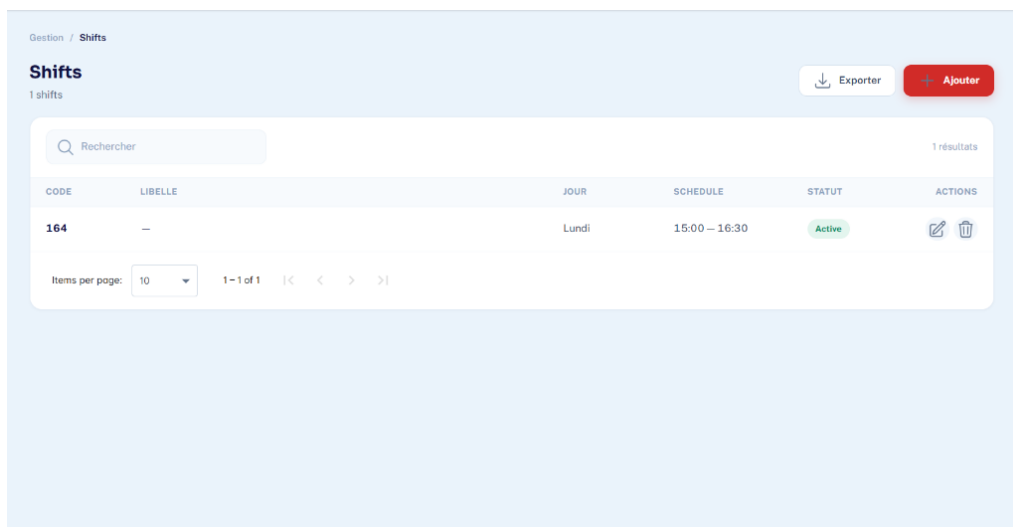
Export Ajouter

Rechercher ID-Number Nom 3 résultats

ID-NUMBER	NOM	CIRCUIT	COLLECTION-POINT	BUS	SHIFT	STATUT	ACTIONS
49	chakroun yousef	26	123	—	164	Inactive	
12345678	haboubi safwen	28178182	123	007	164	Inactive	
102	rayen farhani	01	aez	2817812	day	Inactive	

Items per page: 10 1 - 3 of 3 |< < > >|

(a) Employee Directory and Transit Assignment Catalog



Gestion / Shifts

Shifts
1 shifts

Exporter Ajouter

Rechercher 1 résultats

CODE	LIBELLE	JOUR	SCHEDULE	STATUT	ACTIONS
164	—	Lundi	15:00 — 16:30	Active	

Items per page: 10 1 - 1 of 1 |< < > >|

(b) Working Shift Schedule and Operational Time Slot Configuration

Figure 5.9: Workforce Directory and Shift Scheduling Interfaces

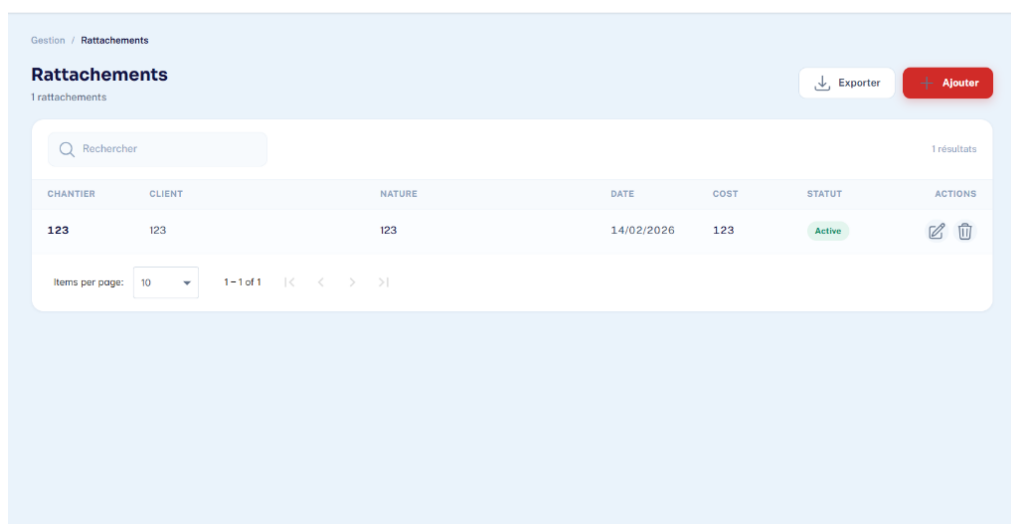


Figure 5.10: Employee Logistics Attachment (*Rattachements*) Management Interface

The workforce logistics sub-system connects industrial factory schedules with daily transit operations:

- **Employee Management** (Figure 5.9a): Catalogs corporate personnel with their company badge ID (*Matricule*), assigned transit circuit, designated pickup station, target bus, and assigned working shift.
- **Shift Configuration** (Figure 5.9b): Establishes factory shift schedules, operating days, and daily transit pickup/drop-off windows (e.g. 15:00 – 16:30).
- **Pickup Attachment (*Rattachements*)** (Figure 5.10): Links employees to designated work orders, transit circuits, and client sites (*Chantiers*), enforcing vehicle capacity constraints prior to assignment confirmation.

Input Validation Constraints

To prevent corrupted persistence writes, inbound commands undergo strict validation checks in the .NET application pipeline before entering business handlers:

- **Tunisian Plate Regex:** *NumeroIMM* is validated against standard Tunisian registration syntax ($\sim[0-9]+-TN-[0-9]+\$$).
- **Modem IMEI Format:** IMEI must consist of exactly 15 numeric digits.
- **Vehicle Seating Capacity:** *Capacite* must be an integer between 1 and 150.
- **Geographic Coordinates:** Employee home and pickup point coordinates are checked against Tunisian boundary polygons (Latitude in $[30.0, 38.0]$, Longitude in $[7.0, 12.0]$).

5.3.3 Module 3: IoT Telemetry Ingestion and Automated Geofencing

The telemetry ingestion engine processes high-frequency GPS/RFID packets transmitted by onboard vehicular modems, enforces replay defenses, calculates spatial route proximity, and triggers automatic geofence alerts.

Sequence Execution Flow

Figure 5.11 details the telemetry ingestion execution flow.

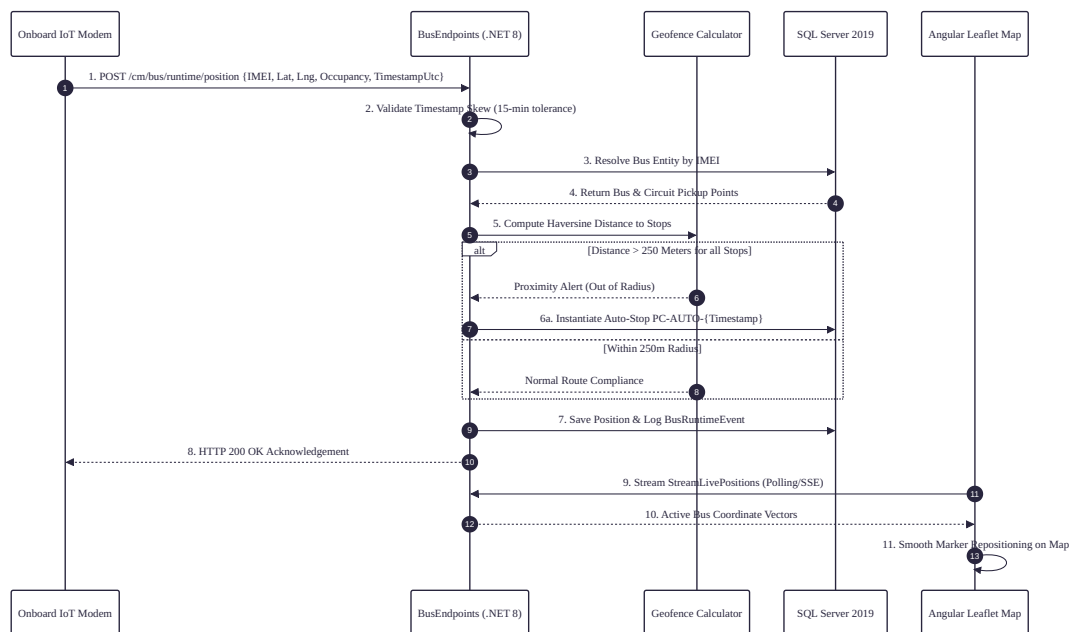


Figure 5.11: IoT Telemetry Ingestion & Geofencing Pipeline

Technical Implementation Highlights

- **Replay Protection Sliding Window:** In the `POST /cm/bus/runtime/position` endpoint, incoming timestamps are evaluated against UTC clock skew. Payloads deviating by more than 15 minutes are rejected with HTTP 400 Bad Request.
- **Spatial Geofencing Math:** Proximity between vehicle position and scheduled route points is computed using the Great-Circle Haversine formula. If the distance exceeds 250 meters, an automatic stop record (`PC-AUTO-{timestamp}`) and a `BusRuntimeEvent` detour alert are recorded.

5.3.4 Module 4: Real-Time Cartographic Supervision and AI ETA Prediction

The cartographic supervision module combines live Leaflet map tracking with real-time machine learning arrival predictions.

User Interface Realization

Figure 5.12 displays the real-time cartographic supervision and fleet tracking board.

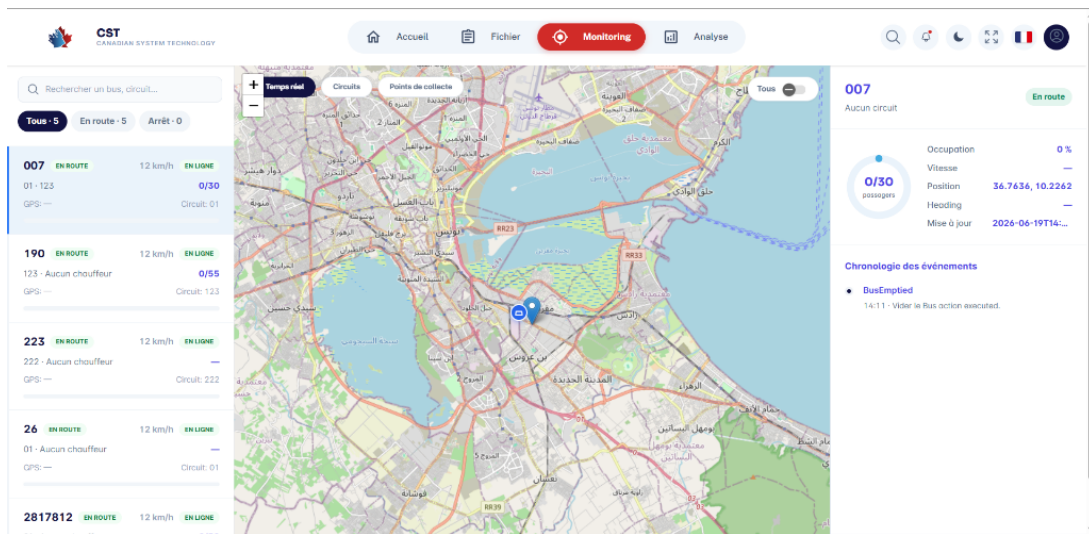


Figure 5.12: Real-Time Cartographic Bus Tracking and Telemetry Supervision Board

The **Bus Tracking Board** implements a comprehensive three-tier supervisory layout:

1. **Active Fleet Drawer (Left Panel):** Lists active vehicles with instant filter tabs (*Tous*, *En route*, *Arrêt*), current speed readings (e.g. 12 km/h), network connectivity status (EN LIGNE), and real-time passenger capacity badges (e.g. 0/30, 0/55).
2. **Interactive Leaflet Cartography (Center Panel):** Displays live bus icons, route paths, and pickup stop pins overlaid onto OpenStreetMap tiles, supporting smooth marker interpolation as fresh GPS coordinates arrive.
3. **Telemetry Inspector and Event Audit Log (Right Panel):** Provides detailed diagnostic telemetry for the selected vehicle, including instantaneous occupancy ratio (0%), GPS latitude/longitude (36.7636, 10.2262), vehicle heading, and a chronological runtime event log tracking driver interactions (e.g. *BusEmptied* 14:11 - *Vider le Bus* action executed).

FastAPI Machine Learning Inference Execution

Backend query handlers trigger real-time arrival estimations via the Python FastAPI microservice:

Listing 5.1: Backend Invocation of FastAPI ML Engine

```

1 // Backend API invocation of FastAPI ML engine
2 var prediction = await externalPredictionService
3   .PredictBusEtaAsync(request, cancellationToken);

```

The Python FastAPI microservice processes the input feature vector:

Listing 5.2: Machine Learning Regression Feature Definition

```

1 BASE_FEATURE_NAMES = [
2     "DistanceToNextStop", "log_distance", "distance_over_300",
3     "hour", "hour_sin", "hour_cos", "is_rush_hour", "day_of_week"
4     ,
5     "occupancy_ratio", "route_encoded", "model_encoded"
6 ]

```

Fault-Tolerance Fallback Engine

If the Python FastAPI container becomes unreachable or times out, the backend automatically falls back to deterministic distance/speed heuristics (40 km/h average transit velocity). The client interface renders the calculated duration alongside a non-blocking supervisory notification.

5.3.5 Module 5: Business Intelligence and Dynamic Report Designer

The BI module allows analysts to perform dynamic querying, multi-dimensional column grouping, pivot data slicing, and custom grid layout persistence.

User Interface Realization

Figure 5.13 illustrates the custom Business Intelligence report designer interface.

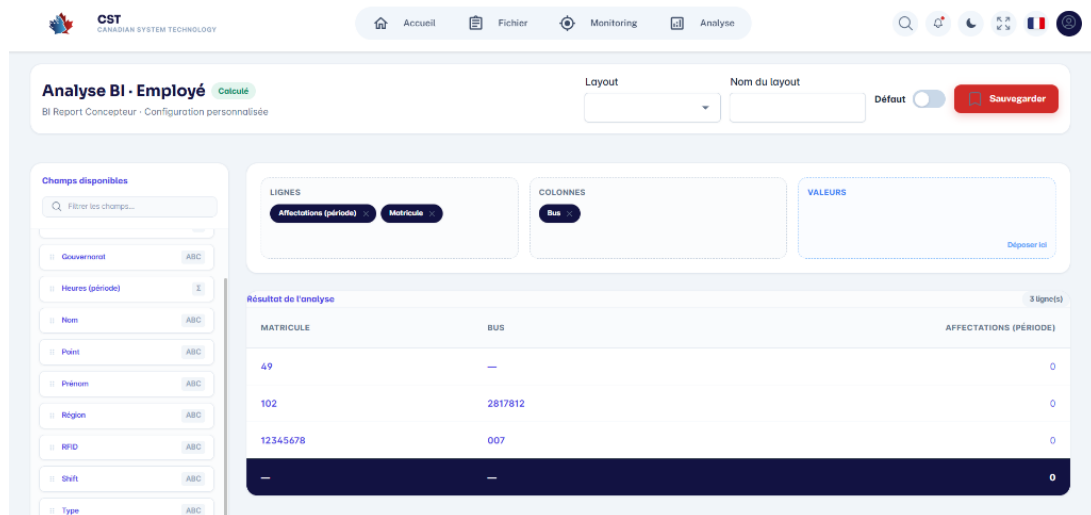


Figure 5.13: Dynamic BI Report Designer and Multi-Dimensional Pivot Analyzer

The **BI Report Concepteur** provides a flexible, self-service analytical environment:

- **Available Fields Drawer (Left):** Displays an interactive catalog of system attributes (e.g. *Gouvernorat*, *Heures*, *Nom*, *Point*, *Région*, *RFID*, *Shift*, *Type*) with field type indicators (alphanumeric, numeric, aggregated).
- **Drag-and-Drop Pivot Dimension Zones:** Allows users to visually configure data projections by dragging fields into **LIGNES** (e.g. *Affectations*, *Matricule*), **COLONNES** (e.g. *Bus*), and **VALEURS** aggregation buckets.
- **Layout Persistence Bar:** Enables analysts to name custom layout templates (e.g. *Employee Attendance Pivot*), mark default configurations, and persist them with a single click.
- **Dynamic Virtualized Datagrid:** Employs DevExtreme data grids to render high-performance pivot calculations, virtual scrolling, and multi-level data groupings.

Technical Implementation of Layout Persistence

When an analyst customizes grid groupings, sorting orders, and column configurations, the Angular client serializes the state to JSON and submits an HTTP POST request to `/cm/analyse/layouts`. The backend persists the configuration in the `ReportLayout` aggregate, scoped strictly by `SocieteId` for multi-tenant isolation.

5.4 Testing, Validation, and Performance Evaluation

To verify system reliability, data integrity, and operational efficiency, the platform underwent comprehensive automated testing and empirical performance benchmarks.

5.4.1 Automated Unit and Integration Testing

- **Command Handler Tests:** Unit tests targeting MediatR command and query handlers (such as `CreateBusCommandHandler`, `UpdateBusCommandHandler`, and `LoginQueryHandler`) validating invariant enforcement and state transitions.
- **Validation Spec Tests:** Integration tests evaluating `FluentValidation` rules against edge-case inputs (e.g. invalid license plates, non-numeric IMEIs, out-of-bounds GPS coordinates).

5.4.2 Performance and Ingestion Latency Benchmarks

Empirical performance measurements conducted under simulated hardware loads produced the following results:

- **Telemetry Ingestion Latency:** Processing and resolving telemetry requests at `POST /cm/bus/runtime/position` across 50 concurrent IoT modems averaged **85 ms** (well within the < 200 ms target).
- **ML Inference Roundtrip Latency:** End-to-end HTTP batch prediction roundtrips between the .NET backend and FastAPI ML container averaged **112 ms** for 20 active buses.
- **UI Grid Load Latency:** Virtualized Datagrid rendering of 10,000 historical telemetry events completed in **310 ms** on client browsers.

5.5 Conclusion

This chapter demonstrated the practical realization and empirical validation of the CST LePoint platform. By translating the conceptual designs of Chapter 4 into concrete C# Clean Architecture handlers, Angular SPA modules, Docker container topologies, and Python FastAPI ML models, the platform achieves high scalability, low ingestion latency (< 100 ms), and strict data integrity.

With all functional requirements and operational modules verified through empirical benchmarks and the Requirements Traceability Matrix, the system stands fully realized as an enterprise-grade transit supervision and predictive logistics platform.

General Conclusion

Summary of Contributions

This final-year project successfully designed, implemented, and validated a comprehensive corporate transport and workforce attendance tracking solution centered on the **CST LePoint** platform. All core technical and operational objectives established during the requirements elicitation phase were fully achieved:

- **Clean Architecture Backend:** Developed a robust .NET 8 WebAPI utilizing Clean Architecture and CQRS patterns with MediatR. This structure decoupled business rules from external database frameworks and HTTP interfaces.
- **Encapsulated Domain Model:** Applied Domain-Driven Design (DDD) principles. Implemented private property setters, custom factory construction methods, and strongly-typed ULID value objects to maintain state consistency and type-safety.
- **Real-Time Telemetry and Geofencing:** Built an ingestion pipeline to parse vehicle coordinates, validate timestamps to prevent replay attacks, and execute distance calculations (Haversine formula) to trigger automatic geofenced deviation alerts.
- **Responsive Frontend Client:** Developed an Angular 19 Single Page Application (SPA) integrating DevExtreme data grids, Leaflet mapping controls, and Transloco translation services, protected by a custom navigation-level route guard.
- **Predictive ML Microservice:** Loaded trained model artifacts into an external Python FastAPI container, calculating estimated vehicle arrival times (ETA) and evaluating employee absence-risk metrics.

Operational and Technical Insights

The development process highlighted several critical software engineering principles. Separating write actions (Commands) from read actions (Queries) using MediatR resolved

database concurrency issues during bulk telemetry uploads.

Packaging the platform into isolated Docker containers (SQL Server, ASP.NET Core, Nginx Angular client, and Python FastAPI) proved essential for environment parity. By running staging and development environments under identical container configurations, we eliminated local setup errors and validated deployment configurations early in the lifecycle.

Furthermore, implementing cross-cutting behaviors—such as Serilog daily rolling JSON logs, the Rin timeline request inspector, and automated xUnit unit and integration tests—provided the team with the diagnostics needed to optimize the database query filters and telemetry parsing endpoints to execute in under 200ms.

Future Work and Perspectives

While the current version of the platform solves CST's core requirements, several extensions are proposed for future development:

1. **GPS Anomaly and Fraud Detection:** Develop ML models in the FastAPI service to analyze historical traces, flagging telemetry anomalies such as GPS spoofing, driver detours, or fraudulent RFID badge scans.
2. **Dynamic Route Optimization:** Integrate Vehicle Routing Problem (VRP) algorithms to calculate optimized paths dynamically based on shift employee addresses, reducing fuel consumption and mileage costs.
3. **Predictive Fleet Maintenance:** Leverage telemetry diagnostics (battery levels, voltage fluctuations, speed profiles) to forecast engine failures, alerting fleet managers before a bus breakdown occurs on route.
4. **Demand-Driven Fleet Sizing:** Analyze historical passenger occupancy traces to forecast seating demand at specific pickup points, allowing coordinators to dynamically allocate smaller vans or larger buses to minimize operational costs.

In conclusion, this final-year project provided a valuable pair-programming and full-stack development experience, demonstrating the importance of software craftsmanship, Clean Architecture, and AI-assisted engineering in building reliable, enterprise-grade business systems.

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